

**FORECASTING COVID-19 WITH
TEMPORAL HIERARCHIES AND ENSEMBLE METHODS**

A Thesis Presented

by

LI SHANDROSS

Submitted to the Graduate School of the
University of Massachusetts Amherst in partial fulfillment
of the requirements for the degree of

MASTER OF SCIENCE

May 2023

Biostatistics and Epidemiology

FORECASTING COVID-19 WITH TEMPORAL HIERARCHIES AND ENSEMBLE METHODS

A Thesis Presented

by

LI SHANDROSS

Approved as to style and content by:

Nicholas G. Reich, Chair

Evan L. Ray, Member

Benjamin W. Rogers, Member

Lisa Chasan-Taber, Department Chair
Biostatistics and Epidemiology

ABSTRACT

FORECASTING COVID-19 WITH TEMPORAL HIERARCHIES AND ENSEMBLE METHODS

MAY 2023

LI SHANDROSS

B.Sc., UNIVERSITY OF MASSACHUSETTS AMHERST

M.S., UNIVERSITY OF MASSACHUSETTS AMHERST

Directed by: Professor Nicholas G. Reich

Infectious disease forecasting efforts underwent rapid growth during the COVID-19 pandemic, providing guidance for pandemic response and about potential future trends. Yet despite their importance, short-term forecasting models often struggled to produce accurate real-time predictions of a complex and rapidly changing system. This gap in accuracy, which persists three years into the pandemic, warrants the exploration and testing of new methods to glean fresh insights.

In this work, we examine the application of the temporal hierarchical forecasting (THieF) methodology to probabilistic forecasts of COVID-19 incident hospital admissions in the United States. THieF is a recent, innovative forecasting technique that aggregates time-series data into a hierarchy made up of different temporal scales, produces forecasts at each level of the hierarchy, then reconciles those forecasts using optimized weighted forecast combination. While THieF’s unique approach has shown substantial accuracy improvements in a diverse range of applications, such as opera-

tions management and emergency room admission predictions, this technique has not previously been applied to outbreak forecasting.

We generate candidate models formulated using the THieF methodology, which differ by their hierarchy schemes and data transformations, and weighted ensembles of the THieF models. The models are validated using weighted interval score (WIS), and the top-performing subset is compared to several benchmark models. These models include simple SARIMA models, a naive baseline model, and a median ensemble of operational incident hospitalization models from the US COVID-19 Forecast Hub. The THieF models and THieF ensembles demonstrate improvements in WIS and MAE, as well as competitive prediction interval coverage, over the benchmark models for both the validation and testing phases. These accuracy improvements suggest the THieF methodology may serve as a promising new technique for infectious disease forecasting that makes better predictions in real-time.

TABLE OF CONTENTS

	Page
ABSTRACT	iii
LIST OF TABLES	vii
LIST OF FIGURES	x
 CHAPTER	
1. INTRODUCTION	1
1.1 Infectious disease forecasting	1
1.2 COVID-19 forecasting	1
1.2.1 US COVID-19 Forecast Hub	1
1.3 Temporal hierarchical forecasting	1
1.4 Goals	1
2. METHODS	2
2.1 Surveillance data	2
2.2 Forecasts	3
2.2.1 Forecast horizons	4
2.2.2 Forecast locations	4
2.2.3 Forecast quantiles	5
2.3 Date range of analysis	6
2.4 Benchmark comparison models	6
2.5 Temporal hierarchical forecasting (THieF)	8
2.5.1 Data Aggregation and Hierarchy	10
2.5.2 Base Forecasts	12
2.5.3 Reconciled Forecasts	12
2.6 Model specifications	12

2.6.1	THieF models	12
2.6.2	THieF ensemble models	13
2.6.3	SARIMA models	15
2.7	Metrics and Evaluation	16
2.7.1	Validation phase	18
2.7.2	Testing phase	19
3.	RESULTS	22
3.1	Validation phase	22
3.2	Testing phase	23
3.2.1	Overall model performance	23
3.2.2	Model performance by horizon week	26
3.2.3	Model performance by pandemic wave	27
3.2.4	Model performance by forecast week	29
3.2.5	Model performance by location	31
4.	DISCUSSION	37
 APPENDICES		
	APPENDIX: SUPPLEMENTAL MATERIALS	38
	 BIBLIOGRAPHY	45

LIST OF TABLES

Table	Page
2.1 Table of the 23 quantiles required by the Forecast Hub and their associated prediction interval bound.	5
3.1 Summary of overall model performance averaged over the states level. A similar ordering of models is seen, with the THieF-4wk-4root model performing the best and the THieF-12wk models performing the worst. However, the THieF-8wk models perform better in terms of model ranking for the states level. Additionally, relative WIS sees smaller values and 50% and 95% PI coverage see values closer to the nominal levels compared to that at the US national level, although relative MAE is larger with every model performing worse than the baseline.	24
3.2 Summary of overall model performance for the US national level, aggregated across forecast week and 1 to 4 week ahead horizons. Prediction interval (PI) coverage rates measure the fraction of times the fraction of times the 50% and 95% PIs covered the true values. A well-calibrated model should achieve close to nominal (0.5 and 0.95, respectively) coverage rates. The “rwis” and “rmae” show relative weighted interval score and relative mean absolute error, which compare each model to the baseline. The baseline is defined to have relative metrics of 1. Models that perform better than the baseline have relative WIS or MAE less than 1. The table is ordered by ascending WIS value. Thus two THieF-4wk models show the best performance out of the THieF models, meeting the baseline for both relative WIS and MAE.	35
3.3 Summary of overall model performance averaged over the states level. A similar ordering of models is seen, with the THieF-4wk-4root model performing the best and the THieF-12wk models performing the worst. However, the THieF-8wk models perform better in terms of model ranking for the states level. Additionally, relative WIS sees smaller values and 50% and 95% PI coverage see values closer to the nominal levels compared to that at the US national level, although relative MAE is larger with every model performing worse than the baseline.	36

3.4	Summary of overall model performance for the US national level, aggregated across forecast week and 1 to 4 week ahead horizons. Prediction interval (PI) coverage rates measure the fraction of times the fraction of times the 50% and 95% PIs covered the true values. A well-calibrated model should achieve close to nominal (0.5 and 0.95, respectively) coverage rates. The “rwis” and “rmae” show relative weighted interval score and relative mean absolute error, which compare each model to the baseline. The baseline is defined to have relative metrics of 1. Models that perform better than the baseline have relative WIS or MAE less than 1. The table is ordered by ascending WIS value. Thus two THieF-4wk models show the best performance out of the THieF models, meeting the baseline for both relative WIS and MAE.	36
1	Summary of overall model performance averaged over the states level. A similar ordering of models is seen, with the THieF-4wk-4root model performing the best and the THieF-12wk models performing the worst. However, the THieF-8wk models perform better in terms of model ranking for the states level. Additionally, relative WIS sees smaller values and 50% and 95% PI coverage see values closer to the nominal levels compared to that at the US national level, although relative MAE is larger with every model performing worse than the baseline.	39
2	Summary of overall model performance for the US national level, aggregated across forecast week and 1 to 4 week ahead horizons. Prediction interval (PI) coverage rates measure the fraction of times the fraction of times the 50% and 95% PIs covered the true values. A well-calibrated model should achieve close to nominal (0.5 and 0.95, respectively) coverage rates. The “rwis” and “rmae” show relative weighted interval score and relative mean absolute error, which compare each model to the baseline. The baseline is defined to have relative metrics of 1. Models that perform better than the baseline have relative WIS or MAE less than 1. The table is ordered by ascending WIS value. Thus two THieF-4wk models show the best performance out of the THieF models, meeting the baseline for both relative WIS and MAE.	40

3	Summary of model performance stratified by horizon week for the states level, aggregated across forecast week. The results from this table were used to create Figure 4 that summarized average WIS by horizon week for the US national level and the states level. The table is ordered by ascending horizon week and WIS, with THieF-4wk-4root having the lowest WIS except for the 3 week (although it is second lowest) and 4 week ahead horizon. The baseline again consistently does not have the best WIS, performing progressively worse compared to the other models for each additional week ahead horizon.	41
4	Summary of model performance stratified by horizon week for the US national level, aggregated across forecast week. The results from this table were used to create Figure 4 that summarized average WIS by horizon week for the US national level and the states level. The table is ordered by ascending horizon week and WIS, with THieF-4wk-4root having the lowest WIS except for the 4 week ahead horizon. The baseline consistently does not have the best WIS.	42
6	Summary of model performance stratified by horizon week and pandemic waves for the states level, aggregated across forecast week. The results from this table were used to create Figure 5 that summarized average WIS by horizon week and pandemic wave for the US national level and the states level. The table is ordered by ascending pandemic wave, horizon week, and WIS, with THieF-4wk-4root again having the lowest WIS for all horizons of the alpha wave and shorter horizons of the Winter 21 wave. The THieF-12wk-noTrans has the lowest WIS for 3 week ahead and 4 week ahead horizons during the Winter 21 wave. The baseline only has the best WIS for the 1 week ahead horizon during the Winter 21 wave.	43
8	Summary of model performance stratified by horizon week and pandemic waves for the US national level, aggregated across forecast week. The results from this table were used to create Figure 5 that summarized average WIS by horizon week and pandemic wave for the US national level and the states level. The table is ordered by ascending pandemic wave, horizon week, and WIS, with THieF-4wk-4root having the lowest WIS for all horizons of the alpha wave and shorter horizons of the Winter 21 wave. The THieF-12wk-noTrans and THieF-12wk-4root models have the lowest and second lowest WIS for 3 week ahead and 4 week ahead horizons during the Winter 21 wave. The baseline consistently does not have the best WIS.....	44

LIST OF FIGURES

Figure		Page
2.1	Overview of the period of analysis included in the paper, superimposed over COVID-19 incident hospitalizations for the entire United States. Vertical dashed lines indicate divisions between phases. The training phase begins on Monday, July 27, 2020 and ends on Monday, July 5, 2021. This phase covers 343 days, or 49 weeks. Forecasts for the training phase are made starting on Monday, December 7, 2020. The testing phase, although not included in this paper, begins Tuesday, July 6, 2021 and ends Monday, November 1, 2021. The division is based on including two waves (the Winter 2020-21 and the Alpha variant) of the pandemic in the training phase and one (Delta variant) in the testing phase. July 27, 2020 marks the beginning of when the forecast locations of interest had reliable truth data.....	7
3.1	U.S. national incident hospitalization forecasts during the testing phase, faceted by model	25
3.2	Average WIS by the horizon week for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week compared to being aggregated across horizon weeks. All models show worse performance relative to the baseline at the 1-week ahead horizon, likely a result of being overconfident at this horizon. Once again, the THieF-4wk-4root model performs the best for WIS. Plots for average MAE are not shown due to their similarity to the average WIS plots.....	28

3.3	Average WIS by the horizon week and pandemic wave for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week and pandemic wave compared to being stratified only across horizon week. Again, all models show worse performance relative to the baseline at the 1-week ahead horizon likely due to being overconfident at this horizon. Model performance is more similar among the top four models during the Alpha wave compared to during the Winter21 wave. Notably, although the THieF-4wk-4root models largely show the lowest WIS, longer horizons during the Winter21 wave saw better performance by the poor-performing THieF-12wk models. Plots for average MAE are not shown due to their similarity to the average WIS plots.	30
3.4	Average h-week ahead WIS and 95% PI coverage for each model for the states level. Once again, model performance differs from week to week with the lowest WIS values during weeks with stable and low incident hospitalization values and the THieF-4wk models showing the best performance. However, the THieF-12wk did not show good performance during high incidence and changed points for the states level; instead, THieF-12-4root actually showed some strangely large WIS values during April and May 2021. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. One-week ahead coverage rates generally improved as time went on, the best models gaining stable and fairly good coverage rates starting in February 2021. For the 4-week ahead horizon, models without a data transformation showed the best performance and stable coverage starting around February 2021, though THieF-4wk-4root and THieF-8wk-4root also saw decent and fairly stable coverage rates after the downward Winter 21 change point.	32

3.5	Average h-week ahead WIS and 95% PI coverage for each model for the US national level. Model performance differs from week to week: the WIS is the lowest during weeks with stable and low incident hospitalization values. The THieF-4wk models show the best performance with lower or comparable WIS to the baseline during the majority of the training period with especially good performance for 1-week ahead horizons. Meanwhile, the THieF-12wk showed lower or comparable WIS to the baseline during high incidence and changed points, which have been historically found difficult to forecast accurately. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. All THieF models show very variable coverage rates, especially for a 1 week ahead horizon, only the models without a data transformation showing stable coverage starting around February 2021.	33
3.6	Relative WIS by location for each model across all horizons for the entire testing phase. Each square contains the average relative WIS or a model at a particular location, colored based on the relative WIS compare to the baseline. Blue indicates model outperformed the baseline for that location on average, while red indicates the model performed worse than the baseline. Locations are sorted based on cumulative hospitalizations during the testing phase while models are sorted from lowest to highest overall relative WIS, aggregated across states (Table 2). The top three models by relative WIS overall during the testing phase outperformed the baseline in every location.	34

CHAPTER 1

INTRODUCTION

- 1.1 Infectious disease forecasting**
- 1.2 COVID-19 forecasting**
 - 1.2.1 US COVID-19 Forecast Hub**
- 1.3 Temporal hierarchical forecasting**
- 1.4 Goals**

CHAPTER 2

METHODS

2.1 Surveillance data

This analysis uses confirmed hospital admissions from HealthData.gov from the COVID-19 Reported Patient Impact and Hospital Capacity by State Time Series. This dataset reports state-aggregated daily incident hospitalizations starting on January 1, 2020 for U.S. states and territories. The data come from three primary sources: HHSTeleTracking, state/territorial health departments which report to HHS Protect, and National Health Care Safety Network (prior to July 15, 2020) [14].

On December 1, 2020, the U.S. COVID-19 Forecast Hub designated this dataset as the official ground truth data for COVID-19 incident hospitalizations. (No other sources were designated official ground truth data prior to this date.) [4] This meant that the HealthData.gov data was used to evaluate the performance of real-time incident hospitalization forecasts submitted to the Forecast Hub. Hence, teams were encouraged to use this hospitalization data set to train their models [5]. In the analysis, we also use this dataset as both input training data for our models and truth data against which to perform accuracy evaluations.

Hospitalization reporting was sporadic for the first four months of the pandemic, with consistent, reliable values for major states and territories achieved only in late July 2020. Specifically, reports of daily confirmed hospitalizations for all 50 states, Washington D.C., Puerto Rico, and the U.S. as a whole began on July 27, 2020. Although COVID-19 hospitalization reporting stabilized after that for cases and deaths, hospitalization values are often more reliable due to requirements for Medicare reim-

bursement. COVID-19 cases have been vastly underreported for the entire duration of the pandemic [7], and this problem has worsened with dwindling testing. Further, truth data collection and reporting for COVID-19 incident cases and deaths has been discontinued by Johns Hopkins University (the official ground truth data source) as of March 10, 2023 [6].

Surveillance data, like the HealthData.gov confirmed hospitalizations, sometimes required revisions due to reporting anomalies. Hospitalizations might be entered on incorrect dates or added after COVID-19 was determined to be the cause of hospitalization. Updates to values usually occurred within the first week or so of posting. As a result, real-time forecast models might train on data that include incident hospitalization numbers for location-date combinations that do not reflect the finalized values.

We query these confirmed hospitalizations from the ‘covidData’ package through the ‘covidHubUtils’ package, both developed by Reich Lab at the University of Massachusetts Amherst to deal with COVID-19 surveillance data. Versions of this data are stored such that users may access the truth data as of a specified date. For example, it is possible to retrieve of version of the confirmed hospitalization data set as of July 27, 2020. To ensure fair comparisons between the models developed for this analysis and some real-time models, our models are trained on the confirmed hospitalizations that would have been available as of the “forecast date” of the retrospective forecast.

2.2 Forecasts

A forecast for a target-location combination (e.g. “1-day ahead incident hospitalizations in Massachusetts”) is defined by one or both of a point forecast and a probabilistic forecast, represented by a set of 23 quantiles [5]. (More information is given about forecast targets, horizons, locations, and quantiles in Sections .) Accord-

ing to conventions set by the Forecasts Hub, all forecast values are truncated to be non-zero, as there is no real-world meaning for negative incident hospitalizations [2]. Further, while this is a retrospective analysis, forecasts are made on Sunday weekly (as they would be if made in real-time) using the data available as of that day. We chose to make forecasts on Sunday because confirmed hospitalization values for subsequent days did not update until Tuesday or later until some time in March 2021, which presented several issues. First, if we began validation forecasting after March 2021, we would lose three months of the period of analysis. Second, the main comparison models made forecasts on Monday, meaning that any models that forecast the week before would be add a disadvantage while any models that forecast on Tuesday or later would be evaluated with the following week’s forecasts. Third, THieF does not allow for missing data points.

2.2.1 Forecast horizons

Short-term forecasts for COVID-19 have been defined to encapsulate horizons of 1- to 4-weeks ahead, or, equivalently, 1- to 28-days ahead [5]. Forecasts for incident hospitalizations are made on a daily time scale, so we choose to restrict¹ the forecast horizons we evaluate to 1- to 28-days ahead.² Horizons are related to target end dates in that a target end date of a particular forecast may be calculated by adding its horizon to the forecast date.

2.2.2 Forecast locations

We make forecasts for 53 of 55 locations from the HealthData.gov confirmed hospitalizations dataset during the period of analysis. These 55 locations include the 50 states, four jurisdictions/territories (Washington D.C., Puerto Rico, American Samoa,

¹Some models make up to 84-day ahead forecasts due to the THieF methodology construction.

²Note, there is a slight exception to this restriction. See Section 2.7 for more information.

and Guam), and a U.S. national level. American Samoa and Guam are excluded due to their very low or zero COVID-19 incident hospitalizations over the entire pandemic. Forecasts for low-count locations that predicted single-digit or zero hospitalizations were often very accurate; hence, these locations do not offer meaningful contributions to our understanding of differences in forecast skill between modeling approaches [5]. Further, even though we include the U.S. national level, the remaining 52 locations are our primary focus, as the magnitude of national forecasts is much greater than out of any single state.

2.2.3 Forecast quantiles

The Forecast Hub specifies the usage of 23 quantiles

$$\mathbb{Q} = \{.010, 0.025, .050, .100, \dots, .900, .950, .990\}$$

to represent a full probabilistic distribution, with the median 0.500 quantile usually taken as the model’s point forecast. We follow this construction, though with a slight modification: prediction intervals are used to define a full distribution of forecasts. The bounds of these intervals can be used to obtain the desired quantiles, as shown in Table 2.2.3 below. For example, the values for a 0.500 and a 0.950 quantile (see the Quantile row) are obtained by taking the lower and upper bounds of the 95% prediction interval (see the Interval row, where "L" and "U" represent the lower and upper bounds, respectively).

Quantile	.010	.025	.050	0.100	0.150	...	0.500	...	0.900	0.950	.975	.990
Interval	98L	95L	90L	80L	70L	...	Median	...	80U	90U	95U	98U

Table 2.1. Table of the 23 quantiles required by the Forecast Hub and their associated prediction interval bound.

2.3 Date range of analysis

We focused on the period between Monday, July 27, 2020 and Sunday, October 2, 2022. This date range spans 798 days, or 114 weeks, which is just over 26 months. The entire period of analysis is split into two main phases: a validation phase of 462 days (66 weeks) and a testing phase of 236 days (48 weeks). The split between these two phases is such that the validation phase ends on Sunday, October 31, 2021 and the testing phase begins on Monday, November 1, 2021.

The first day of the period of analysis (July 27, 2020) was selected as the first day of reliable truth data for all locations of interests. Forecasting for the validation phase began on Monday, December 7, 2020, as this is the first Monday after the Health-Data.gov confirmed hospitalizations data set was declared the official ground truth data. The division between the validation and testing phases was chosen to encapsulate three COVID-19 variant-based waves in the validation phase (Winter 2020-21, Alpha, and Delta) and two waves in the testing phase (Omicron and BA.4/BA.5). The separation between pandemic waves is made based on U.S. national level incident hospitalizations, pictured below in Figure ??.

2.4 Benchmark comparison models

Two models from the Forecast Hub—the COVIDhub-baseline and COVIDhub-4_week_ensemble—are selected to serve as comparisons for the models in this analysis during the testing phase evaluation. Many of the metrics used in this evaluation are not especially meaningful on their own, the value often dependent on the magnitude of the observed data, requiring a relative comparison among models forecasting the same targets (1- to 28-day ahead incident hospitalizations for COVID-19) [5].

The COVIDhub-baseline is a naive baseline model created and run by the US COVID-19 Forecast Hub, designed as a simple benchmark for comparison. The baseline makes forecasts using the previous day’s incident hospitalization value as the

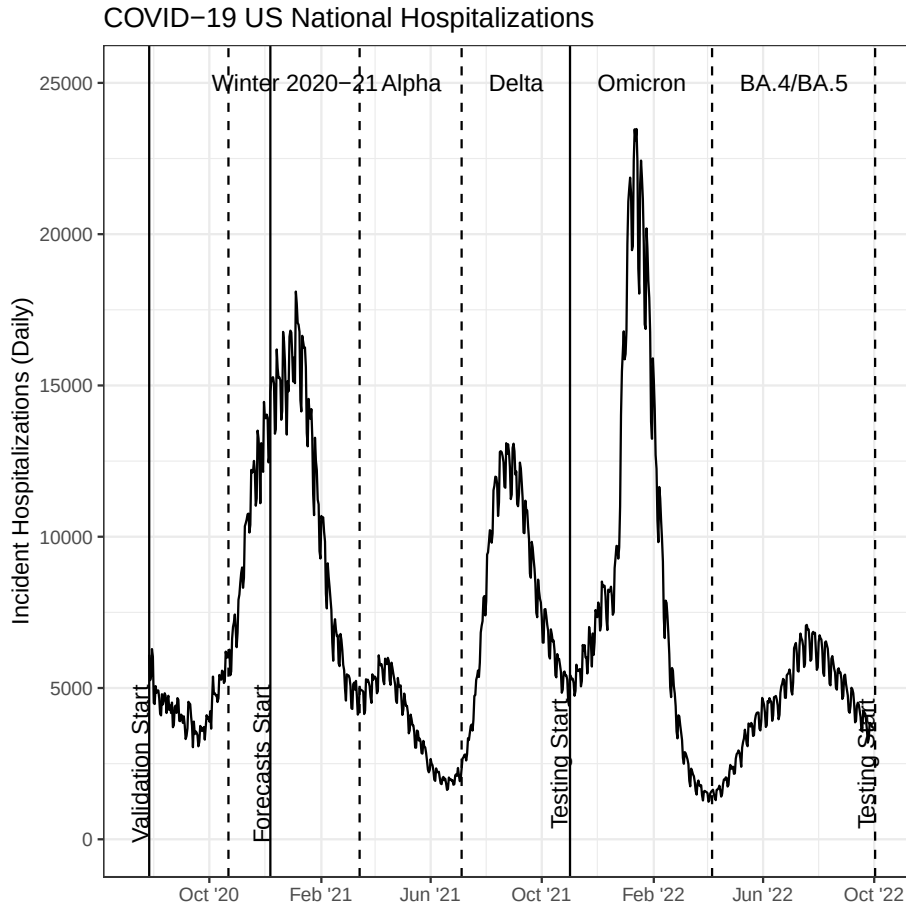


Figure 2.1. Overview of the period of analysis included in the paper, superimposed over COVID-19 incident hospitalizations for the entire United States. Vertical dashed lines indicate divisions between phases. The training phase begins on Monday, July 27, 2020 and ends on Monday, July 5, 2021. This phase covers 343 days, or 49 weeks. Forecasts for the training phase are made starting on Monday, December 7, 2020. The testing phase, although not included in this paper, begins Tuesday, July 6, 2021 and ends Monday, November 1, 2021. The division is based on including two waves (the Winter 2020-21 and the Alpha variant) of the pandemic in the training phase and one (Delta variant) in the testing phase. July 27, 2020 marks the beginning of when the forecast locations of interest had reliable truth data.

point forecast (and 0.5 quantile for the probabilistic forecast) and using past changes in daily incidence to calibrate uncertainty in its probabilistic forecasts [5]. Beating the baseline is typically seen as one marker of a useful model.

The COVIDhub-4_week_ensemble is an equally-weighted median ensemble of eligible³ submitting models for incident hospitalizations at the COVID-19 Forecast Hub, run by the Hub team that includes forecasts up to 28-day (4-week) ahead horizons. This model produces the same 1- to 14-day (1- to 2-week) ahead horizon forecasts as the COVIDhub-ensemble model, with the addition longer horizons [2]. These 15- to 28-day (3- to 4-week) horizons were removed from the main ensemble model, as research showed a sharp decline in accuracy in forecasts predicting beyond 2-week aheads [12]. However, we still include such longer horizons in the analysis for comparison with our models.

Throughout period of analysis, the COVIDhub-4_week_ensemble consisted of a shifting list of contributing models due to changing eligibility criteria and submitting models for incident hospitalizations [2]. This may have had an impact on shifting accuracy of the ensemble, though the extent is unclear.

The forecasts for the two comparison models are obtained from the Zoltar forecast archive, a research data repository developed by Reich Lab, through the ‘covidHubUtils’ package [15].

2.5 Temporal hierarchical forecasting (THieF)

Historically, hierarchical forecasting has been utilized in business, economics, operations management, and similar fields. This type of forecasting brings advantages like producing coherent forecasts⁴ and increasing prediction accuracy compared to

³For more information about forecast eligibility, see [2]

⁴Hierarchical forecasts are coherent if the forecasts at lower levels sum up to those at higher levels. Coherence is not an intrinsic property of all hierarchical forecasts. A lack of coherence causes

other methods [1]. Forecast combination has likewise shown favorable outcomes in a myriad of applications. The unique merging of forecast combination with hierarchical forecasting in the THieF methodology has resulted in promising accuracy improvements over more traditional techniques that solely involve hierarchical forecasting [1].

Forecast combination that integrates forecasts from different, independent modeling frameworks to create ensembles often results in better infectious disease forecasts compared to those from individual models [11]. Likewise, classical statistical models have demonstrated good performance for short-term forecasts in outbreak settings [9]. THieF’s built-in forecast combination (which shares some key similarities with that of ensemble methods) and underlying ARIMA base-forecaster suggest that this methodology will show accuracy improvements when forecasting COVID-19 incident hospitalizations.

Surveillance data tends to be quite noisy due to reporting artifacts, especially when considering small time scales COVID-19 incident hospitalizations suffer a day-of-the-week effect as a result of hospital and clinic operating hours. Fewer hospitalizations are reported on Saturday and Sunday, instead often recorded as part of Monday hospitalizations, since patients are not admitted on weekends. We may better capture true short-term trends by aggregating daily hospitalizations to be weekly or every two weeks, although such smoothing carries some risk of excluding useful information at the daily level. Thus, we are careful to include both daily and weekly aggregation levels in the THieF temporal hierarchy construction for every model. Other higher aggregation levels are included in model schemes to investigate the existence of hidden patterns at higher time scales and if such a pattern could improve model accuracy.

a dilemma in which it is hard to know which level of forecasts to trust, but a single level must be chosen regardless [8].

The THieF methodology consists of three steps to produce forecasts from time-series data using a time-based hierarchical structure [1]. We begin with truth data aggregation to create the temporal hierarchy. The daily confirmed hospitalizations dataset is a time-series consisting of $T = 798$ observations that can be represented as $\{y_t; t = 1, \dots, 798\}$. These data are then aggregated to different time scales based on the aggregation levels of interest for a particular model (e.g. weekly, every two weeks, every four weeks, etc.⁵). While each aggregation level can be thought of as a separate, univariate time series with different frequencies, together they constitute the temporal hierarchy for one model. Next, we make base forecasts at every level of the hierarchy using an ARIMA model for each level. These forecasts will likely not be coherent, meaning that they predict different trends for the next $h = 28$ days. This disagreement between the forecasts is desirable at this stage, as it may indicate that important patterns have been detected at different time scales. However, through the step of reconciliation (making the forecasts coherent), the information from all of the aggregation levels are combined into forecasts that are expected to be more accurate than any single level. Following this stage, the forecasts from every timescale will agree about the amount of daily incident hospitalizations for the next 1 to 28 days.

2.5.1 Data Aggregation and Hierarchy

Suppose we are interested in constructing a THieF model that combines forecasts from 1-week, 2-week, and 4-week intervals because we believe these time scales each contain important information. Aggregate series for each of these times scales can be constructed from the original daily time series. In fact, any combination of time

⁵see Section 2.6.1 for a full list of time scales considered

scales can be chosen for the hierarchy as long as they are factors⁶ of the topmost aggregation level (see Section 2.6.1 for some examples).

m is the number of observations of the smallest timescale (or highest sampling frequency) that aggregate to create a single data point for the highest level. In our example case, $m = 28$ since there are 28 days in 4 weeks. $k \in \{k_p, \cdot, k_2, k_1\}$ can take on multiple values and indicates the particular aggregation level, where $k_p = m$, $k_1 = 1$, and p is the total number of aggregation levels. In our example case, $k \in \{28, 14, 7\}$, where $p = 3$ Athanasopoulos et. al provide a mathematical equation to obtain the data points $y_j^{[k]}$ in each aggregated series, written as

$$y_j^{[k]} = \sum_{t=t^*+(j-1)k}^{t^*+jk-1} y_t$$

for $j = 1, \dots, \lfloor T/k \rfloor$, $M_k = m/k$ for the aggregated series. We define $t^* = T - \lfloor T/m \rfloor m + 1$ to ensure non-overlapping aggregation using our example, we can calculate $t^* = 798 - \lfloor 798/28 \rfloor 28 + 1 = 15$, then we obtain the first data point for the 1-week time series with the equation $y_1^{[7]} = \sum_{t=15+(0)7}^{15+7-1} y_t = y_{15} + \dots + y_{21}$, which is as we expect given that the first t^* observations are removed. However, the index j used above is clearly aggregation-level specific, with no way to translate between levels of the hierarchy. Hence, the authors propose an alternative index i where i is the observation index of the most aggregate series $y_i^{[m]}$ so $j = i$ at that level. A more general form for all levels is $y_j^{[k]} = y_{M_k(i-1)+z}^{[k]}$ for $z = 1, \dots, M_k$. This new index i is used for the remaining equations and in the figure below. Using our example, we can show that the new notation is consistent with the old notation: $y_5^{[7]} = y_{M_7(1)+1}^{[7]} = y_{4+1}^{[7]}$. For each period of the top-level aggregation (2 weeks in our example), we create a column vector of data points

⁶Non-factor levels tend to be more difficult to compute and do not necessarily abide by the following equations.

Each aggregation level can be thought of as a univariate time series from which forecasts can be made. Next, a classical statistical model is used to make these base forecasts. These consist of forecasts of up to 84-day ahead incident hospitalizations, depending on the forecasting model (although only up to 28-day ahead forecasts are evaluated).

2.5.2 Base Forecasts

any given date's 1- to 28-day ahead base forecasts represented as $\{\hat{y}_h; h = 1, \dots, 28\}$

2.5.3 Reconciled Forecasts

2.6 Model specifications

We create three distinct classes of models for comparison during the validation phase: the original THieF models, the THieF ensembles, and the SARIMA models. In total, these three groups consist of twenty-six (26) candidate models.

2.6.1 THieF models

The THieF models are constructed using the THieF methodology outlined in Section 2.5 with different combinations of aggregation levels and data transformations. Seven hierarchical structures and two data transformations result in fourteen (14) original THieF models. Each model is defined (and named) based on the highest level of its temporal hierarchy and the data transformation performed on the forecasts. For example, the simplest of the THieF models is THieF_1wk-noTransform while the most complicated is THieF_12wk-4root.

The aggregation levels of any model's temporal hierarchy are daily, plus f_m -week periods where f_m is the set of all factors for the highest time scale m . Hence, the THieF_12wk-4root model (as well as the THieF_12wk-noTransform model) has a temporal hierarchy made up of daily, weekly, 2-weekly, 3-weekly, 4-weekly, 6-weekly,

and 12-weekly aggregation levels since $f_{12} = \{1, 2, 3, 4, 6, 12\}$. Meanwhile, the THieF_1wk-noTransform model’s temporal hierarchy is only made up of daily and weekly aggregation levels ($f_1 = \{1\}$). We chose 1-week, 2-week, 3-week, 4-week, 6-week, 8-week, and 12-week top-level aggregations so as to have a range of time scales to inform the reconciled forecasts.

We explore the seven hierarchy schemes with both no data transformation and a fourth-root data transformation. The latter transform is implemented by first taking the fourth root of the confirmed hospitalizations time series, using the truth data to make forecasts with the THieF methodology, then performing an inverse fourth-power transformation to the resulting forecasts. We selected the fourth-root transformation based on prior research that had shown that had shown accuracy improvements for COVID-19 forecasting due to its variance-stabilizing effect [13].

2.6.2 THieF ensemble models

The THieF ensemble models⁷ are seven mean ensembles of the fourteen total original THieF models. This is true even during the testing phase during which not all fourteen original THieF models make forecasts (see Section 2.7.2).

Of the seven THieF ensembles, we formulated six as trained ensembles, in which past performance of the component THieF models is used to assign the components weights, and one as an untrained ensemble, in which all component forecasters are assigned equal weight [10]. Both types of ensembles make quantile forecasts⁸ at level k by combining those of the component forecasts for a particular forecasts date, location, horizon, and quantile combination. (For example, a particular model m might make a 0.500 quantile forecast on December 7, 2020 that predicts the number

⁷The equations used to define the ensembles are adapted from Ray, et. al [10]

⁸Recall that we use the 0.500 quantile forecast as the point forecast.

of 1-day ahead incident hospitalizations in Massachusetts will be 185.) An ensemble quantile forecast can be expressed mathematically as

$$q_{l,s,h,k}^{ens} = f(q_{l,s,h,k}^1, \dots, q_{l,s,h,k}^M),$$

where l is the location, s is the forecast date, h is the horizon, and the superscript $m = 1, \dots, M$ denotes the model [10].

A trained, weighted mean ensemble calculates its forecast quantiles as

$$q_{l,s,h,k}^{ens} = \sum_{m=1}^M w_s^m q_{l,s,h,k}^m$$

, in which the weights are calculated using a sigmoidal transformation of the component models' relative WIS over a rolling window of twelve (12) weeks leading up to the ensemble's forecast date s :

$$w_s^m = \frac{\exp(-\theta \cdot rWIS_s^m)}{\sum_{m'=1}^M \exp(-\theta \cdot rWIS_s^{m'})}.$$

When the non-negative parameter $\theta = 0$, this becomes an equally weighted ensemble. However, as θ increases, the weights become more unequal, with the contributions of better-performing component models eclipsing that of worse-performing models. Note that the weights, however, are always non-negative and sum to one [10].

The THieF ensembles are named with the θ value used in their construction, except for the untrained ensemble, which is called THieF_ensemble-mean. We select θ values $\{0, 1, 3, 6.5, 10, 15, 20, 25\}$ to encompass the range of values taken on by the COVIDhub-trained_ensemble⁹ when predicting incident hospitalization [10]. The one

⁹A trained ensemble created by the Forecast Hub that makes short-term forecasts for incident cases, incident and cumulative deaths, and incident hospitalizations. This model makes its forecasts by computing a weighted median of the ten best component forecasts (by WIS) in the 12 weeks leading up to the forecast date. For this model, θ is allowed to vary by forecast date [2, 10].

non-integer value (6.5) was selected as the median (and approximate mean) θ value over the period of analysis for all locations of interest except for the U.S. national level. The corresponding ensemble was named THieF_ensemble-train6.5.

While the value of θ generally remains static for each ensemble, all trained ensembles are defined to start with $\theta = 0$ until 12 weeks of prior forecasts are achieved (on March 1, 2021) during the validation phase that can be used to calculate weights based on prior performance of forecasts. During the testing phase, the value of θ is dependent on at least some validation phase performance until the thirteenth week (beginning on January 24, 2022).

2.6.3 SARIMA models

The four SARIMA (seasonal ARIMA) models are classic statistical models created using the ‘auto.arima()’ function from the ‘forecast’ package in ‘R’. This function makes forecasts based on input time series data, and the user may specify seasonality when creating a SARIMA model. Using the original HealthData.Gov confirmed hospitalizations data set, we create two ARIMA models and two SARIMA models, with either no data transformation or a variance-stabilizing fourth root transformation, as described in THieF Models. The SARIMA models both have a seasonality of 7 days (1 week), named sarima_s7-4root and sarima_s7-noTransform, while the ARIMA models have no seasonality, named sarima_s1-4root and sarima_s1-4root.

The construction of the SARIMA models is intentionally identical to that of the daily and weekly aggregation levels of the original THieF models, though in practice the resulting base forecasts from the THieF models may differ slightly due to the aggregation process to create the temporal hierarchy. We specifically want to use the SARIMA models to help investigate whether it is the THieF methodology’s temporal hierarchy and forecast reconciliation that impacts its performance, compared to

simply the underlying ARIMA base forecaster. (Recall that research has shown that classic statistical models have shown good performance for outbreak forecasting [9].)

2.7 Metrics and Evaluation

Evaluation of our models is conducted by aggregating the output forecasts in seven ways: by forecast date, by location, by pandemic wave, overall validation/testing phase, and in the last 17 weeks (of the validation phase). Despite being made on a daily scale, hospitalization forecasts have been evaluated on a weekly scale by the Forecast Hub, and we follow this convention. For each metric, a single value is obtained by averaging the values across a single week (Tuesday through Monday) [3]. However, modeling teams have made forecasts on different days (e.g. Saturday vs Sunday vs Monday), leading to a mismatch between what day an h -day ahead forecast is predicting for. Hence, for the models we construct, we actually use 3- to 30-day ahead horizons to ensure all forecasts are predicting for the same dates. It should be noted, however, this may result in a slight disadvantage for models whose 3- to 30-day ahead forecasts are being compared to 1- to 28-day ahead forecasts.

We use the same metrics as the Forecast Hub to evaluate model accuracy for both the validation and testing phases. Average mean absolute error (MAE) is used to score point forecasts, while average weighted interval score (WIS), average 50% prediction interval (PI), and average 95% PI coverage are used to score probabilistic forecasts. (We refer to these metrics simply as MAE, WIS, and 50% and 95% PI coverage going forward for simplicity.) Averages for these metrics are first calculated for probabilistic forecasts (and point forecasts) for a single location, horizon, and quantile combination, then the mean is taken for each model from forecasts aggregated in one of the seven ways mentioned above. The resulting values from the four metrics are adapted from Cramer, et. al [5] and are given as follows:

1. $MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$

$$2. WIS_{\alpha_0:K}(F, y_i) = \frac{1}{N} \sum_{i=1}^N \frac{1}{K+1/2} (w_0 \cdot |y_i - m| + \sum_{k=1}^K w_k \cdot IS_{\alpha_k}(F, y_i))$$

$$3. (1 - \alpha) * 100\% \text{ PI Coverage} = \frac{1}{N} \sum_{i=1}^N 1(l_i^\alpha \leq y_i \leq u_i^\alpha),$$

where N is the number of forecasts being averaged, y_i is the true observed value at a particular time point (date) and location, and $\hat{y}_i$ is the point forecast value for a particular forecast date, horizon, and location combination. The PI uncertainty level is given by α , k is the particular prediction interval ($k = 1, \dots, K$), and F is the forecast distribution. The weights w_0 and w_k are set as $w_0 = 1/2$ and $w_k = \alpha_k/2$ respectively while u_i^α and l_i^α are the upper and lower bounds of a $(1 - \alpha) * 100$ -level prediction interval for a particular forecast date and location. Lastly, the interval score of a particular $(1 - \alpha) * 100$ -level prediction interval is $IS_{\alpha_k}(F, y_i) = (u_i - l_i) + \frac{2}{\alpha} \cdot (l_i - y_i) \cdot 1(y_i < l_i) + \frac{2}{\alpha} \cdot (y_i - u_i) \cdot 1(y_i > u_i)$ where $1()$ is the indicator function.

Relative versions of MAE and WIS are also used to adjust for truth data magnitude to allow for comparison between forecast values of different scales. These relative metrics are calculated as $MAE_{\text{model m}}/MAE_{\text{baseline}}$ and $WIS_{\text{model m}}/WIS_{\text{baseline}}$, respectively.

Evaluation for each of the seven aggregation categories (except for individual locations) are separated by location category: U.S. national and averaged states. The truth data and resulting forecasts are of different magnitudes for the aggregated U.S. national and average states levels. This means that the contribution from the U.S. national location would account for an overwhelming proportion of the resulting forecast scores if we did not make such a split.

We perform no statistical tests to assess the significance of performance differences between models, as certain issues arise when making time series forecasts that prevent usage of such tests. Namely, scores tend to be highly-correlated across multiple dimensions, like horizons, forecast dates, locations, etc. Assumptions of independent, identically distributed data found in standard inference are invalid in time series forecasting. In practice, numerical comparisons of forecast error between prospective

models (without excluding any models that were compared) is sufficient for choosing the best model to inform decision-making, and foregoing formal statistical tests is common practice in infectious disease forecasting. Still, it should be noted that we cannot necessarily make a definitive statement about the predominance of one model over another without formal statistical testing [9].

2.7.1 Validation phase

The validation phase serves as a period during which to train the candidate models, calibrate the forecasts, and to select the best models to pass on to the testing phase based on performance during the validation phase. We opt to call this period the "validation phase" over the more common "training phase" because truth data from this first phase continues to inform the forecasts made during the testing phase, unlike is defined for a more traditional training-testing split of observed data for model validation.

All validation phase evaluations are restricted to forecasts with target end dates (date for which the prediction of incident hospitalizations is being made) within the validation phase (July 27, 2020 to October 31, 2021), with forecasts beginning on Monday, December 7, 2020. We select one model that meets the criteria for each of the following categories to pass onto the testing phase:

- Best original THieF overall for average states location category
- Best original THieF overall for U.S. national location category
- Best THieF trained ensemble overall for average states location category
- Best THieF trained ensemble overall for U.S. national location category
- Best SARIMA overall for average states location category
- Best SARIMA overall for U.S. national location category

- Best original THieF in last 17 weeks for average states location category
- Best original THieF in last 17 weeks for U.S. national location category
- THieF untrained ensemble,

where "best" is defined by lowest WIS.

Based on these criteria, a maximum of nine models will make up the best subset to move onto the training phase evaluations, if unique candidates occupy each category. However, if the same model occupies multiple categories, we may pass on as few as five models.

We stratify by location category due to a difference in magnitude, as described above. For each location category, one model is passed on for each of the model classes—original THieF, THieF (trained) ensemble, SARIMA—as the single best candidate during the entire validation phase. We also select the single best original THieF model during the last 17 weeks of the validation phase, a category designed to account for any models whose performance improves with larger amounts of data on which to train on (since we allow models to use validation phase data to inform their forecasts). This category is limited to THieF models only, as it is models with large aggregation levels that we suspect may be fit this description, and the SARIMA models include only very short seasonality (if any) while the THieF ensemble models incorporate forecasts all the THieF models. The THieF_untrained-ensemble is passed on regardless of performance, as we are interested in comparing the performance of this model to that of the best trained THieF ensemble.

2.7.2 Testing phase

Using the subset of best candidate models found during the validation phase, we make new testing phase forecasts and evaluate these models against both each other and the two comparison models from the Forecast Hub, COVIDhub-baseline

and COVIDhub-4_week_ensemble. Like with the validation phase, all testing phase evaluations are restricted to forecasts with target end dates within the testing phase (November 1, 2021 to October 2, 2022).

The testing phase analysis is broken down into five smaller evaluations, based on aggregating the testing phase forecasts for the overall period, by horizon week, by pandemic wave, by location, and by forecast date, each split by geographic scale (except for by location, where such a split would be redundant). The smaller evaluations are intended to help pinpoint any potential accuracy gains (or losses) the original THieF or THieF ensembles have over the COVIDhub-baseline (or COVIDhub-4_week_ensemble) and allow us to hypothesize the origin of these accuracy differences.

The overall period evaluation provides a generalized view of the testing phase analysis and is meant to identify a single best model that could be used to inform decision-making¹⁰. The by-horizon evaluation allows identification of models that perform well at shorter horizons (typically 1 to 2 week ahead), as the larger WIS and MAE values for longer horizons (3 to 4 week ahead) will dominate these metrics in the overall period evaluation. This dominance is particularly undesirable given that the Forecast Hub ensemble models, which tend to be some of the most accurate models overall, experience a sharp decline in accuracy beyond two week ahead horizons for incident hospitalizations [12].

The by-pandemic wave and by-forecast date evaluations allow investigation into time periods in which the number of incident hospitalizations is of different magnitudes and is changing at different rates. Times of greater values or of rapid changes have larger WIS and MAE and are generally harder to predict [7, 12], and it may be useful to identify which models perform best under different scenarios. These two evaluations are also stratified by horizon week. However, the by-location evaluation,

¹⁰Though of course we cannot say this definitively given that we do not perform any formal statistical tests [9].

which provides insight into differences in model performance between higher and lower count locations, is not stratified by horizon week.

CHAPTER 3

RESULTS

3.1 Validation phase

Tables 3.1 and 3.1 show the overall model ranking and evaluation metrics during the validation phase for the averaged states and U.S. national location scales, in order of ascending WIS. We can see that for the averaged location scale the THieF ensembles generally perform the best, followed by the original THieF models, then the SARIMA models. This pattern is not as strictly followed for the U.S. national location scale, in which the two original THieF models with a 6-week top aggregation level have lower WIS than at least half of the THieF ensembles.

For averaged states, we pass on THieF_6wk-4root, THieF_ensemble-train3, and sarima_s7-noTransform as the best models of each class during the validation phase overall. While THieF_ensemble-train3 is not strictly the highest ranked THieF ensemble for this location scale, we pass it along based on the selection rule from cross-validation in which the simplest model should be chosen if it does not perform substantially worse than more complex models. For U.S. national, we pass along THieF_6wk-noTransform, THieF_ensemble-train3, and sarima_s7-noTransform as the best representative of each class for the overall validation phase of evaluation. The best original THieF models during the last 17 weeks of the validation phase are THieF_6wk-4root and THieF_12wk-4root for averaged states and U.S.national, respectively, with the full evaluation results shown in Supplemental Tables 4 and 4. The untrained THieF ensemble, while not a top performer for either location scale in the overall analysis, is also passed on to serve as a reference model.

Hence, the best subset that passes onto the testing phase consists of the following six (6) models: THieF_6wk-4root, THieF_6wk-noTransform, THieF_ensemble-train3, sarima_s7-noTransform, THieF_12wk-noTransform, and THieF_ensemble-mean.

3.2 Testing phase

Using forecasts made every four weeks during the testing phase, we plot 1- to 28-day ahead incident hospitalization point and 95% interval forecasts for the eight models of interest during this period in Figures ???. This group consists of six models from the best subset and two Forecast Hub comparison models, the COVIDhub-baseline and the COVIDhub-4_week_ensemble. Generally, the THieF ensembles, the original THieF models with 6-week top aggregation levels, and COVIDHub-4_week_ensemble appear to better follow the true confirmed hospitalizations compared to the other three models, and these five models also tend to have narrower 95% interval forecasts. These results are mirrored (and perhaps further explained) by the following stratified evaluations.⁴⁹⁸

3.2.1 Overall model performance

Both location scales show the same model ranking for the overall testing phase evaluation, with the top five models outperforming the COVIDhub-baseline. The COVIDhub-4_week_ensemble takes the lead for best WIS and MAE by a significant margin. The next best three models—THieF_6wk-4root, THieF_ensemble-train3, THieF_ensemble-mean—all have fairly similar relative WIS and relative MAE values, as do the two worst-performing models: THieF_12wk-noTransform and sarima_s7-noTransform. However, there are some notable differences between performance for averaged states versus U.S. national.

For averaged states (see Table ??), the WIS and MAE values are more closely clustered together and are of a smaller magnitude. Every best subset model is well-

	model	wis	mae	cov_50	cov_95	rwis	rmae
1	THieF_ensemble-train10	30.73	47.22	0.53	0.93	0.86	0.94
2	THieF_ensemble-train15	30.74	47.28	0.52	0.93	0.86	0.95
3	THieF_ensemble-train6.5	30.77	47.20	0.53	0.94	0.86	0.94
4	THieF_ensemble-train20	30.77	47.33	0.52	0.93	0.86	0.95
5	THieF_ensemble-train25	30.81	47.37	0.51	0.93	0.86	0.95
6	THieF_ensemble-train3	30.90	47.25	0.54	0.94	0.86	0.95
7	THieF_ensemble-train1	31.33	47.44	0.53	0.94	0.87	0.95
8	THieF_6wk-4root	31.65	48.11	0.46	0.91	0.88	0.96
9	THieF_6wk-noTransform	32.26	48.67	0.54	0.92	0.90	0.97
10	THieF_1wk-4root	32.68	50.06	0.48	0.91	0.91	1.00
11	THieF_2wk-4root	32.94	50.26	0.47	0.91	0.92	1.01
12	THieF_2wk-noTransform	33.26	48.03	0.57	0.90	0.93	0.96
13	THieF_1wk-noTransform	33.74	49.21	0.58	0.90	0.94	0.98
14	THieF_8wk-4root	33.84	51.14	0.43	0.89	0.94	1.02
15	THieF_4wk-4root	34.10	51.50	0.46	0.90	0.95	1.03
16	THieF_ensemble-mean	34.29	47.65	0.53	0.94	0.96	0.95
17	THieF_3wk-4root	34.62	52.71	0.46	0.90	0.96	1.05
18	THieF_8wk-noTransform	34.76	52.06	0.51	0.90	0.97	1.04
19	THieF_3wk-noTransform	34.96	52.08	0.54	0.90	0.97	1.04
20	THieF_4wk-noTransform	35.63	52.82	0.53	0.89	0.99	1.06
21	COVIDhub-baseline	35.88	49.98	0.83	0.98	1.00	1.00
22	sarima_s7-noTransform	36.29	51.91	0.57	0.89	1.01	1.04
23	sarima_s7-4root	36.35	54.55	0.46	0.90	1.01	1.09
24	sarima_s1-4root	36.91	55.43	0.46	0.89	1.03	1.11
25	THieF_12wk-noTransform	39.14	59.06	0.42	0.89	1.09	1.18
26	sarima_s1-noTransform	39.41	56.38	0.54	0.88	1.10	1.13
27	THieF_12wk-4root	84.32	54.49	0.41	0.86	2.35	1.09

Table 3.1. Summary of overall model performance averaged over the states level. A similar ordering of models is seen, with the THieF-4wk-4root model performing the best and the THieF-12wk models performing the worst. However, the THieF-8wk models perform better in terms of model ranking for the states level. Additionally, relative WIS sees smaller values and 50% and 95% PI coverage see values closer to the nominal levels compared to that at the US national level, although relative MAE is larger with every model performing worse than the baseline.

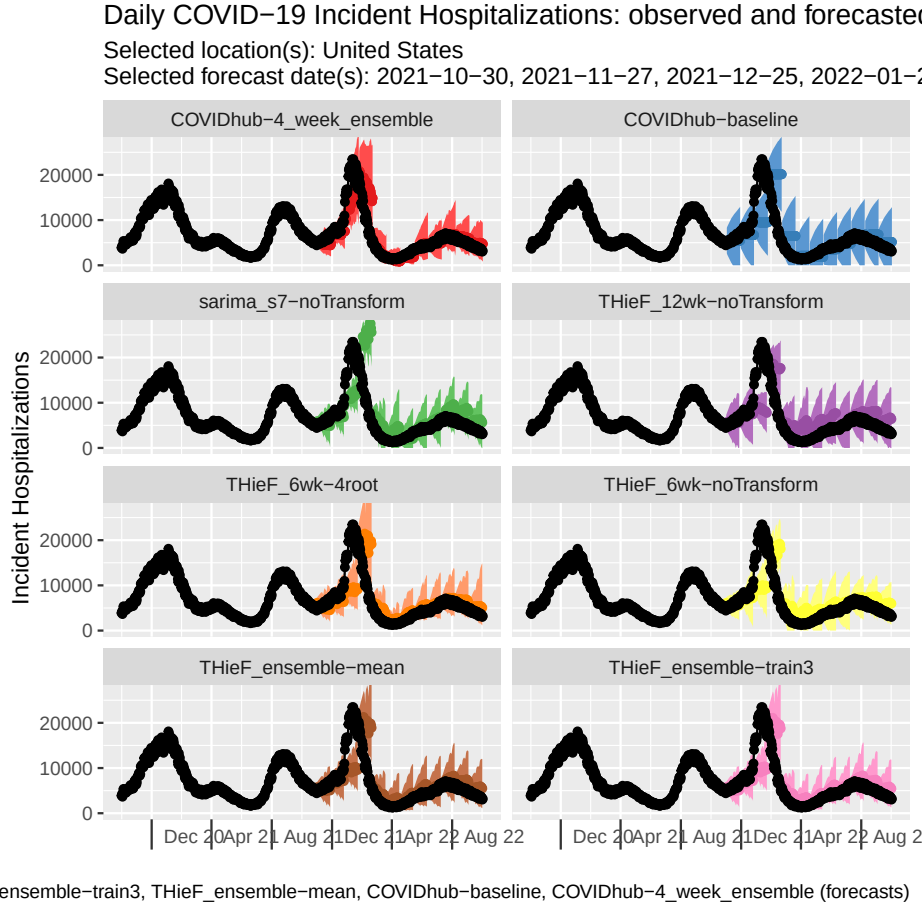


Figure 3.1. U.S. national incident hospitalization forecasts during the testing phase, faceted by model

calibrated, with near-nominal coverage rates at both uncertainty levels. In fact, our models had better 50% coverage rates than those of the Forecast Hub models. In addition, for this location scale the models in the lower half of the ranking, including the COVIDhub-baseline, all showed very similar relative WIS and relative MAE values.

For U.S. national (see Table ??), however, the WIS and MAE values tend to be more spread out, partially a result of their larger magnitude. At this location scale, our models had below-nominal coverage rates for both uncertainty levels. The Forecast Hub models had above-nominal coverage rates at the 50% level but also fell short of nominal coverage at the 95% level. The THieF_6wk-noTransform model showed

more similar performance to the top four ranked models compared to that of the COVIDhub-baseline, unlike for averaged states. The bottom two models for the U.S. national ranking were also substantially worse compared to the COVIDhub-baseline in terms of relative WIS and relative MAE values.

3.2.2 Model performance by horizon week

Only WIS plots are shown because this is our principle evaluation metric and the MAE plots are very similar.

The by-horizon evaluation shows similar model rankings to those of the overall evaluation (see Figure ?? and Supplemental Table ??). The top four models for both location scales are the same and perform no worse (but usually tend to do better) than the COVIDhub-baseline at all horizons, and the COVIDhub-4_week_ensemble significantly outperforms all other seven models. However, aggregating the forecasts by horizon illustrates some important differences in accuracy between horizon weeks.

For averaged states (see Figure ??), THieF_6wk-4root outperforms the THieF ensembles for all horizons, except for 1-week ahead. Similarly, THieF_6wk-noTransform is better than three models at 2- to 4-weeks ahead but is only better than THieF_12wk-noTransform for the 1-week horizon. Further, while sarima_s7-noTransform is ranked fifth and THieF_12-noTransform is ranked last for the 1-week ahead horizon, these ranks flip to be sixth and last for the 3- and 4-week ahead horizons, after briefly crossing with each other (and the COVIDhub-baseline) at 2-weeks ahead. The COVIDhub-baseline is consistently the second worst model.

For U.S. national (see Figure ??), THieF_6wk-4root outperforms the THieF ensembles for the two longest horizons while the ensembles (and COVIDhub-baseline) perform slightly better at the shortest horizon. The difference in performance among the three models is barely discernible at a horizon of 2 weeks. The sarima_s7-noTransform and THieF_12wk-noTransform models occupy the bottom of the rank-

ing, with THieF_12wk-noTransform scoring lower WIS values for all but the 1-week ahead horizon. The difference is especially pronounced at longer horizons, where this model has much lower WIS than that of sarima_s7-noTransform. The COVIDhub-baseline ranks sixth at the U.S. national location scale for all but the 1-week ahead horizon where it ranks fifth.

3.2.3 Model performance by pandemic wave

One overall theme that continues for the by-pandemic wave evaluation (see Figure ?? and Supplemental Tables ??) is that the COVIDhub-4_week_ensemble ranks first for lowest WIS, and this model is substantially better than all the other models at every horizon. For each pandemic wave-location scale combination, most horizons show the next three models to be THieF_6wk-4root and the two THieF ensembles, with THieF_12wk-4root and sarima_s7-noTransform ranking near the bottom. The COVIDhub-baseline also ranks similarly low among the eight models. Plots for the same pandemic wave generally bear greater resemblance in terms of overall trends than plots for the same location scale; the one notable exception is numerical scale for average WIS.

For the Omicron variant wave, the best and worst models—COVIDhub-4_week_ensemble and sarima_s7-noTransform—are easily distinguishable from the middle six models. THieF_ensemble-mean and THieF_ensemble-train3 rank second and third for short horizons but drop to fourth and fifth for the 4-week ahead horizon. THieF_6wk-noTransform performs better at all horizons for the U.S. national scale during this wave. The COVIDhub-baseline also appears to perform better for U.S. national. On the other hand, sarima_s7-noTransform has more comparable performance for the averaged states scale at every horizon. All models have greater WIS values during the Omicron wave compared to during the BA.4/BA.5 wave for their respective location scale.

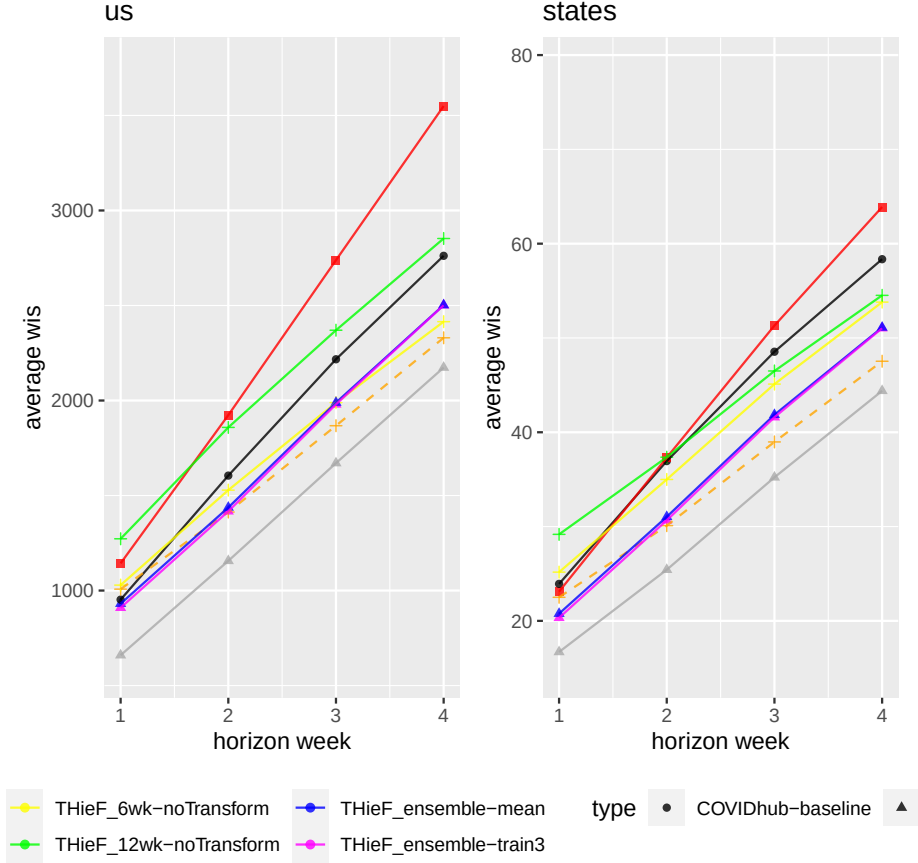


Figure 3.2. Average WIS by the horizon week for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week compared to being aggregated across horizon weeks. All models show worse performance relative to the baseline at the 1-week ahead horizon, likely a result of being overconfident at this horizon. Once again, the THieF-4wk-4root model performs the best for WIS. Plots for average MAE are not shown due to their similarity to the average WIS plots.

For the BA.4/BA.5 variant wave, model rankings are more distinct and tend to remain static from one horizon week to another. The best three and worst models are the same regardless of location scale; however, the COVIDhub-baseline is noticeably better ranked for U.S. national scores, where it is approximately tied with THIEF_ensemble-mean for fourth place, compared to a clear seventh place for averaged states scores. THIEF_6wk-noTransform also performs as well or better than sarima_s7-noTransform for all horizons during this wave at the U.S. national scale, while the ranks of these two models depends on horizon length at the averaged states scale.

3.2.4 Model performance by forecast week

- best to worst: hub ensemble, thief ensembles 6wk-4root sarima baseline 6wk-noTransform 12 wk - models take turns at being best/worst under different scenarios
- 12 wk model noticeably average part way down decline of omicron wave but usually the worst otherwise - 6wk-noTransform best/among best starting part way down omicron decline, performed well during ba4/ba5 wave (often second best); seems like its poorer aggregated performance has to do with omicron peak and first half of decline
- thief ensembles not always the best but perform usually in top half; ranked first just prior to omicron peak once, first or near first several times ba4/ba5wave - trained ensemble just barely beats out mean ensemble for almost all weeks - baseline usually at bottom for stable, low count times - Best to worst: Hub ensemble Thief ensembles, 6wk-4 root 6 WK-noTransform 12 WK, baseline, sarima (but this is a lot harder to determine) - models take turns at being best/worst under different scenarios - 12 week performs quite well (average to top) during times of rapid change during omicron, also second half of incline to ba4/ba5 peek - 6wk-noTransform is similar to 12 week, those slightly better performance for first half of the a 4/ba5 in coin - thief ensembles not always the best but perform usually in top half; best performance is as good as Hub

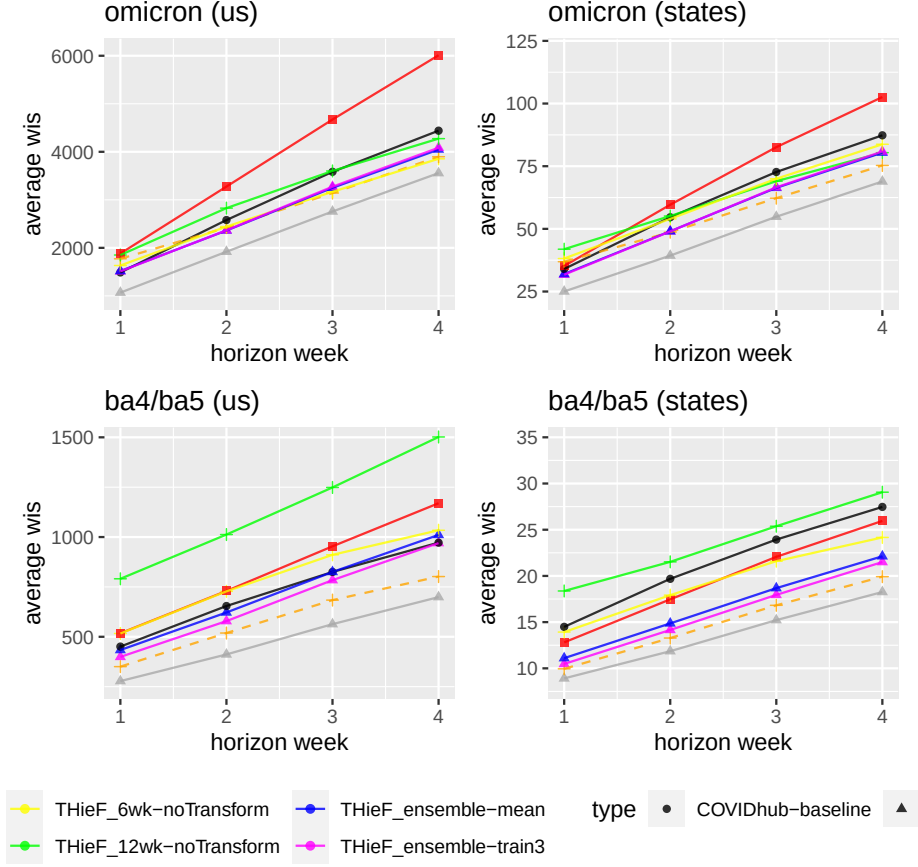


Figure 3.3. Average WIS by the horizon week and pandemic wave for each model for the US national level and the states level. The figures show similar trends in WIS when forecasts are stratified by horizon week and pandemic wave compared to being stratified only across horizon week. Again, all models show worse performance relative to the baseline at the 1-week ahead horizon likely due to being overconfident at this horizon. Model performance is more similar among the top four models during the Alpha wave compared to during the Winter21 wave. Notably, although the THieF-4wk-4root models largely show the lowest WIS, longer horizons during the Winter21 wave saw better performance by the poor-performing THieF-12wk models. Plots for average MAE are not shown due to their similarity to the average WIS plots.

ensemble leading up to ba4/ba5 peak - ensembles alternate a bit more about which one is best, 60:40 train 3 wins - baseline usually at bottom for stable, low count times; only occasionally ranks high - Worst to best: 12 week 6 week-noTransform sarima 6 week-4 root, thief ensembles hub ensemble baseline - sarima has worst coverage part way through omicron decline - at same time, hub ensemble has best coverage - thief ensembles have decent; often are closer to nominal level, especially for stable, low count times - Incredibly low coverage just before omicron peak, all models suffer but especially 6 and 12 week-no transform - Coverage (worst to best): sarima, 6wk-noTransform, 12wk, ensembles, 6wk-4root, hub-ensemble, baseline - notably, baseline, 6wk-4root, 12wk, and thief ensembles beat out hub-ensemble around peak of omicron wave - sarima had comparable coverage (its best) during ba4/ba5wave - 6wk-4root had strangely low coverage following fall of omicron wave

3.2.5 Model performance by location

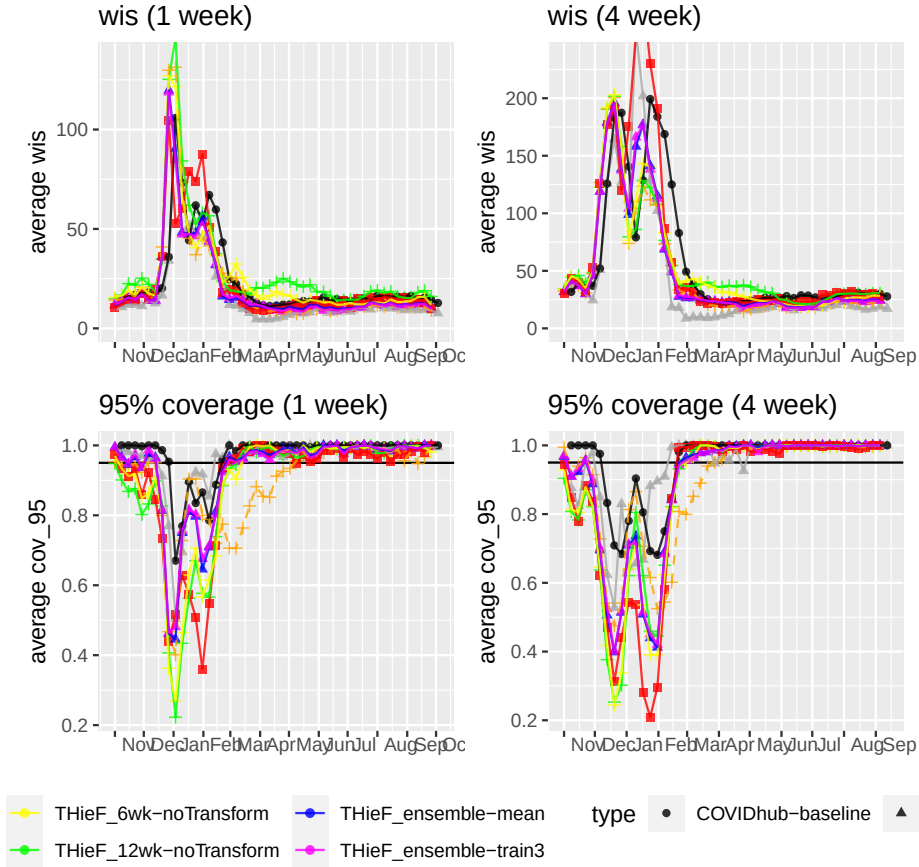


Figure 3.4. Average h-week ahead WIS and 95% PI coverage for each model for the states level. Once again, model performance differs from week to week with the lowest WIS values during weeks with stable and low incident hospitalization values and the THieF-4wk models showing the best performance. However, the THieF-12wk did not show good performance during high incidence and changed points for the states level; instead, THieF-12-4root actually showed some strangely large WIS values during April and May 2021. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. One-week ahead coverage rates generally improved as time went on, the best models gaining stable and fairly good coverage rates starting in February 2021. For the 4-week ahead horizon, models without a data transformation showed the best performance and stable coverage starting around February 2021, though THieF-4wk-4root and THieF-8wk-4root also saw decent and fairly stable coverage rates after the downward Winter 21 change point.

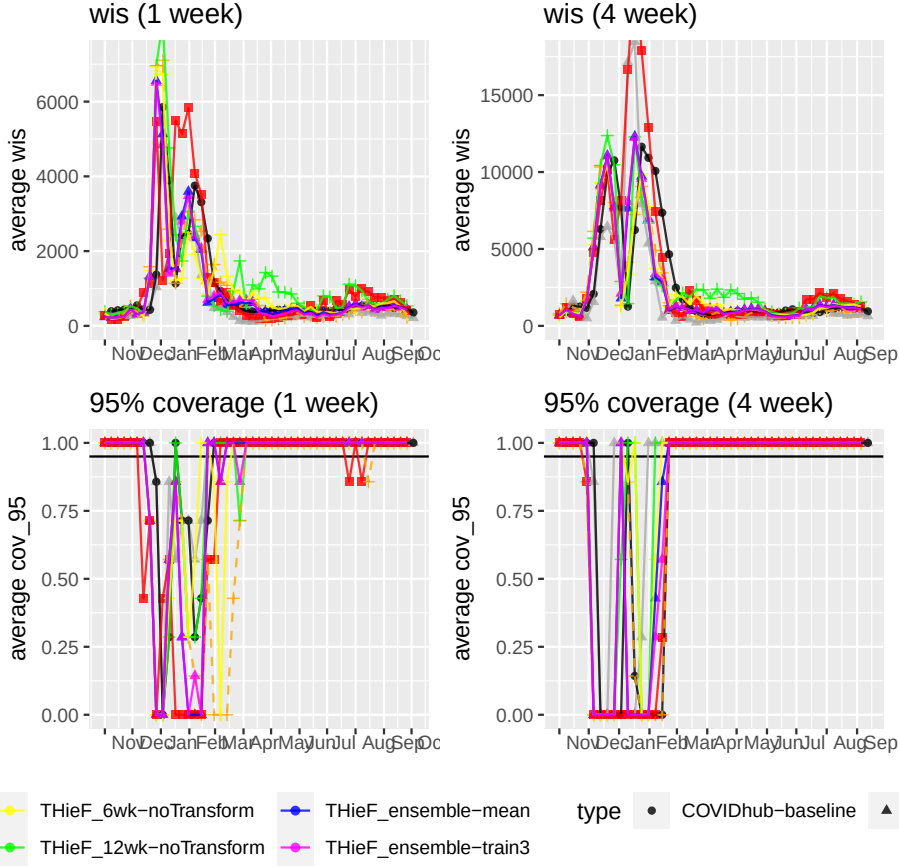


Figure 3.5. Average h-week ahead WIS and 95% PI coverage for each model for the US national level. Model performance differs from week to week: the WIS is the lowest during weeks with stable and low incident hospitalization values. The THieF-4wk models show the best performance with lower or comparable WIS to the baseline during the majority of the training period with especially good performance for 1-week ahead horizons. Meanwhile, the THieF-12wk showed lower or comparable WIS to the baseline during high incidence and changed points, which have been historically found difficult to forecast accurately. Plots for h-week ahead MAE are not shown due to their similarity to the WIS plots. All THieF models show very variable coverage rates, especially for a 1 week ahead horizon, only the models without a data transformation showing stable coverage starting around February 2021.

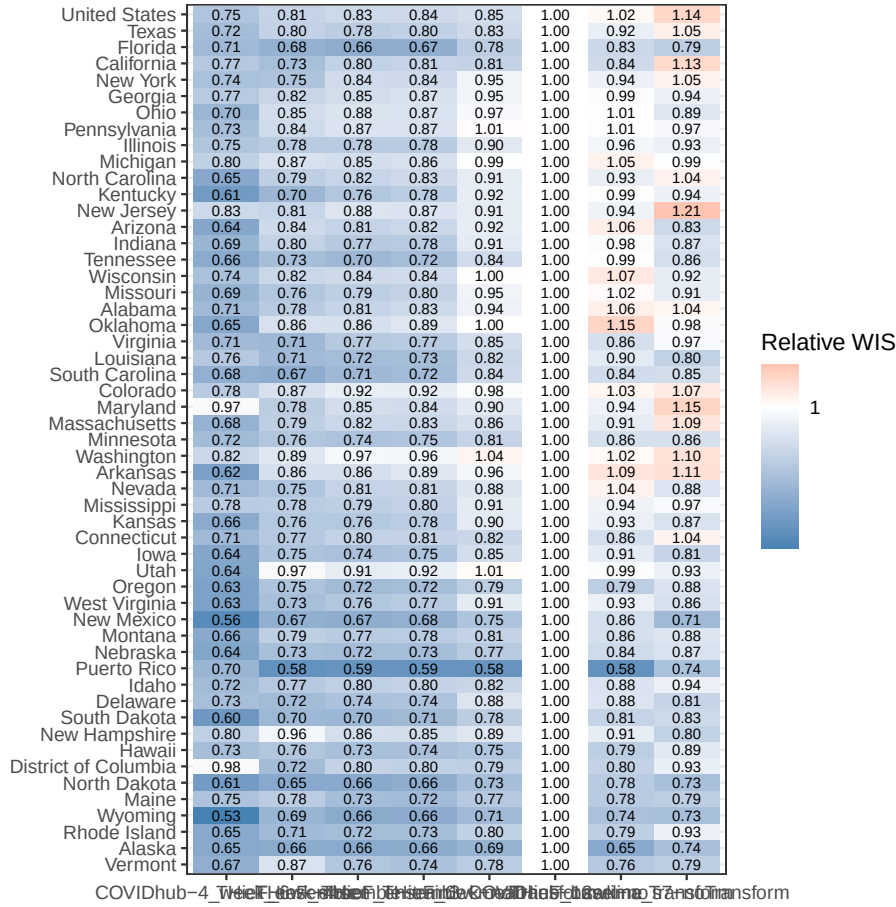


Figure 3.6. Relative WIS by location for each model across all horizons for the entire testing phase. Each square contains the average relative WIS of a model at a particular location, colored based on the relative WIS compare to the baseline. Blue indicates model outperformed the baseline for that location on average, while red indicates the model performed worse than the baseline. Locations are sorted based on cumulative hospitalizations during the testing phase while models are sorted from lowest to highest overall relative WIS, aggregated across states (Table 2). The top three models by relative WIS overall during the testing phase outperformed the baseline in every location.

	model	wis	mae	cov_50	cov_95	rwis	rmae
1	THieF_6wk-noTransform	1116.40	1756.00	0.36	0.91	0.88	0.88
2	THieF_ensemble-train3	1139.80	1808.60	0.34	0.91	0.89	0.91
3	THieF_ensemble-train1	1144.90	1814.90	0.35	0.90	0.90	0.91
4	THieF_ensemble-train6.5	1146.60	1817.40	0.34	0.91	0.90	0.91
5	THieF_6wk-4root	1152.40	1765.80	0.33	0.76	0.90	0.88
6	THieF_ensemble-train10	1159.20	1834.00	0.33	0.91	0.91	0.92
7	THieF_ensemble-train15	1180.20	1866.30	0.33	0.90	0.92	0.94
8	THieF_ensemble-train20	1199.40	1897.50	0.32	0.90	0.94	0.95
9	THieF_ensemble-mean	1212.70	1822.30	0.35	0.90	0.95	0.91
10	THieF_ensemble-train25	1214.30	1919.30	0.32	0.89	0.95	0.96
11	THieF_1wk-noTransform	1218.50	1856.40	0.46	0.87	0.95	0.93
12	THieF_2wk-4root	1234.70	1941.80	0.25	0.85	0.97	0.97
13	THieF_1wk-4root	1252.20	1993.10	0.30	0.85	0.98	1.00
14	COVIDhub-baseline	1276.50	1995.90	0.64	0.99	1.00	1.00
15	THieF_8wk-noTransform	1277.00	1968.90	0.38	0.85	1.00	0.99
16	THieF_8wk-4root	1303.60	1963.80	0.26	0.70	1.02	0.98
17	THieF_2wk-noTransform	1343.10	1985.20	0.40	0.82	1.05	0.99
18	THieF_4wk-4root	1383.40	2058.00	0.28	0.70	1.08	1.03
19	THieF_3wk-noTransform	1392.90	2008.60	0.38	0.78	1.09	1.01
20	THieF_4wk-noTransform	1397.40	2038.40	0.35	0.77	1.09	1.02
21	THieF_3wk-4root	1401.70	2164.80	0.26	0.80	1.10	1.08
22	sarima_s1-noTransform	1477.90	2162.00	0.37	0.79	1.16	1.08
23	sarima_s7-noTransform	1483.20	2200.60	0.39	0.81	1.16	1.10
24	sarima_s1-4root	1530.20	2227.10	0.26	0.75	1.20	1.12
25	THieF_12wk-noTransform	1553.50	2357.80	0.32	0.76	1.22	1.18
26	sarima_s7-4root	1601.50	2467.20	0.22	0.76	1.25	1.24
27	THieF_12wk-4root	2270.60	2039.90	0.34	0.66	1.78	1.02

Table 3.2. Summary of overall model performance for the US national level, aggregated across forecast week and 1 to 4 week ahead horizons. Prediction interval (PI) coverage rates measure the fraction of times the fraction of times the 50% and 95% PIs covered the true values. A well-calibrated model should achieve close to nominal (0.5 and 0.95, respectively) coverage rates. The “rwis” and “rmae” show relative weighted interval score and relative mean absolute error, which compare each model to the baseline. The baseline is defined to have relative metrics of 1. Models that perform better than the baseline have relative WIS or MAE less than 1. The table is ordered by ascending WIS value. Thus two THieF-4wk models show the best performance out of the THieF models, meeting the baseline for both relative WIS and MAE.

	model	wis	mae	cov_50	cov_95	rwis	rmae
1	COVIDhub-4_week_ensemble	30.18	44.87	0.64	0.94	0.72	0.79
2	THieF_6wk-4root	34.55	50.12	0.51	0.90	0.83	0.88
3	THieF_ensemble-train3	35.62	51.57	0.56	0.91	0.86	0.91
4	THieF_ensemble-mean	35.91	51.80	0.57	0.91	0.86	0.91
5	THieF_6wk-noTransform	39.51	56.87	0.55	0.88	0.95	1.00
6	COVIDhub-baseline	41.63	56.77	0.79	0.96	1.00	1.00
7	THieF_12wk-noTransform	41.66	60.97	0.51	0.88	1.00	1.07
8	sarima_s7-noTransform	43.54	58.94	0.60	0.87	1.05	1.04

Table 3.3. Summary of overall model performance averaged over the states level. A similar ordering of models is seen, with the THieF-4wk-4root model performing the best and the THieF-12wk models performing the worst. However, the THieF-8wk models perform better in terms of model ranking for the states level. Additionally, relative WIS sees smaller values and 50% and 95% PI coverage see values closer to the nominal levels compared to that at the US national level, although relative MAE is larger with every model performing worse than the baseline.

	model	wis	mae	cov_50	cov_95	rwis	rmae
1	COVIDhub-4_week_ensemble	1401.30	2085.10	0.63	0.91	0.75	0.83
2	THieF_6wk-4root	1642.50	2358.80	0.40	0.82	0.88	0.94
3	THieF_ensemble-train3	1687.40	2443.40	0.36	0.83	0.90	0.97
4	THieF_ensemble-mean	1700.50	2458.90	0.39	0.84	0.91	0.98
5	THieF_6wk-noTransform	1727.70	2504.80	0.46	0.86	0.92	1.00
6	COVIDhub-baseline	1867.50	2514.10	0.71	0.85	1.00	1.00
7	THieF_12wk-noTransform	2074.10	2908.30	0.37	0.86	1.11	1.16
8	sarima_s7-noTransform	2316.80	3042.40	0.41	0.80	1.24	1.21

Table 3.4. Summary of overall model performance for the US national level, aggregated across forecast week and 1 to 4 week ahead horizons. Prediction interval (PI) coverage rates measure the fraction of times the 50% and 95% PIs covered the true values. A well-calibrated model should achieve close to nominal (0.5 and 0.95, respectively) coverage rates. The “rwis” and “rmae” show relative weighted interval score and relative mean absolute error, which compare each model to the baseline. The baseline is defined to have relative metrics of 1. Models that perform better than the baseline have relative WIS or MAE less than 1. The table is ordered by ascending WIS value. Thus two THieF-4wk models show the best performance out of the THieF models, meeting the baseline for both relative WIS and MAE.

CHAPTER 4

DISCUSSION

- thief ensembles are best - seems like there's no real meaningful difference between mean ensemble and weighted $\langle U \rangle = 3$ - both essentially always beat the baseline for WIS and MAE with all stratified evaluations - but don't quite beat the hub-ensemble - thief_6wk-4root is next best, then thief_6wk-noTransform - 4root is consistently better and more often beats the baseline for non-coverage metrics - noTransform is always as good or better as baseline for non-coverage metrics - sarima and thief_12wk-noTransform are the worst - tend to be worse than the baseline but except in their specialties: - sarima performs very well in static, low-count periods/locations - 12wk performs very well in rapidly changing, high-count periods/locations - but even if they occasionally beat other models (especially thief ensembles), they tend to be only as good or worse - so it seems like ensembling is the way to go to get the best of both worlds

- Limitation of overall metrics being dominated by longer horizons and states with high true hospitalization counts - Future implementation of log scores (maybe to address the above??) - Using applications of thief specifically tailored for probabilistic forecasting, not just point forecasting

APPENDIX: SUPPLEMENTAL MATERIALS

	model	wis	mae	cov_50	cov_95	rwis	rmae
1	THieF_ensemble-train6.5	32.40	49.69	0.52	0.92	0.76	0.77
2	THieF_ensemble-train10	32.41	49.81	0.52	0.93	0.76	0.77
3	THieF_ensemble-train3	32.45	49.55	0.52	0.92	0.76	0.77
4	THieF_ensemble-train15	32.50	50.03	0.51	0.93	0.76	0.78
5	THieF_ensemble-mean	32.53	49.23	0.52	0.92	0.76	0.76
6	THieF_ensemble-train1	32.54	49.49	0.52	0.92	0.76	0.77
7	THieF_ensemble-train20	32.63	50.26	0.51	0.92	0.76	0.78
8	THieF_ensemble-train25	32.77	50.48	0.51	0.92	0.77	0.78
9	THieF_6wk-4root	33.13	49.34	0.50	0.91	0.77	0.77
10	THieF_2wk-4root	33.73	51.33	0.50	0.91	0.79	0.80
11	THieF_2wk-noTransform	34.19	50.15	0.50	0.88	0.80	0.78
12	THieF_6wk-noTransform	34.78	51.56	0.50	0.88	0.81	0.80
13	THieF_1wk-4root	34.85	53.57	0.49	0.91	0.81	0.83
14	THieF_8wk-4root	34.91	51.66	0.48	0.90	0.82	0.80
15	THieF_12wk-4root	34.92	50.48	0.49	0.91	0.82	0.78
16	THieF_8wk-noTransform	35.45	52.57	0.49	0.88	0.83	0.82
17	THieF_3wk-4root	36.07	54.64	0.49	0.90	0.84	0.85
18	THieF_4wk-4root	36.66	54.82	0.48	0.89	0.86	0.85
19	THieF_3wk-noTransform	36.98	54.12	0.48	0.87	0.86	0.84
20	THieF_1wk-noTransform	37.21	54.92	0.47	0.86	0.87	0.85
21	THieF_4wk-noTransform	38.33	56.40	0.47	0.86	0.90	0.88
22	THieF_12wk-noTransform	38.49	53.65	0.49	0.85	0.90	0.83
23	sarima_s1-4root	41.55	62.71	0.44	0.88	0.97	0.97
24	COVIDhub-baseline	42.78	64.41	0.72	0.97	1.00	1.00
25	sarima_s7-4root	43.72	64.94	0.45	0.88	1.02	1.01
26	sarima_s7-noTransform	43.81	62.42	0.45	0.84	1.02	0.97
27	sarima_s1-noTransform	46.51	67.02	0.42	0.83	1.09	1.04

Table 1. Summary of overall model performance averaged over the states level. A similar ordering of models is seen, with the THieF-4wk-4root model performing the best and the THieF-12wk models performing the worst. However, the THieF-8wk models perform better in terms of model ranking for the states level. Additionally, relative WIS sees smaller values and 50% and 95% PI coverage see values closer to the nominal levels compared to that at the US national level, although relative MAE is larger with every model performing worse than the baseline.

	model	wis	mae	cov_50	cov_95	rwis	rmae
1	THieF_12wk-4root	1109.40	1606.60	0.60	0.86	0.71	0.62
2	THieF_ensemble-mean	1147.80	1863.50	0.37	0.91	0.74	0.71
3	THieF_6wk-4root	1155.80	1761.00	0.48	0.88	0.74	0.68
4	THieF_ensemble-train1	1170.20	1909.10	0.37	0.92	0.75	0.73
5	THieF_ensemble-train3	1189.00	1947.60	0.35	0.92	0.76	0.75
6	THieF_6wk-noTransform	1216.00	1905.90	0.33	0.86	0.78	0.73
7	THieF_ensemble-train6.5	1229.20	2013.30	0.34	0.94	0.79	0.77
8	THieF_ensemble-train10	1274.70	2080.70	0.34	0.94	0.82	0.80
9	THieF_8wk-noTransform	1293.30	2063.20	0.35	0.86	0.83	0.79
10	THieF_ensemble-train15	1341.00	2185.10	0.33	0.94	0.86	0.84
11	THieF_12wk-noTransform	1386.60	1952.40	0.55	0.77	0.89	0.75
12	THieF_ensemble-train20	1399.10	2278.40	0.33	0.92	0.90	0.87
13	THieF_8wk-4root	1409.10	2103.70	0.34	0.79	0.91	0.81
14	THieF_1wk-noTransform	1410.30	2230.50	0.28	0.79	0.91	0.86
15	THieF_2wk-4root	1444.60	2287.50	0.28	0.86	0.93	0.88
16	THieF_ensemble-train25	1444.90	2348.90	0.32	0.92	0.93	0.90
17	THieF_4wk-noTransform	1446.60	2130.50	0.34	0.67	0.93	0.82
18	THieF_3wk-noTransform	1465.90	2126.00	0.34	0.66	0.94	0.82
19	THieF_2wk-noTransform	1471.60	2181.30	0.31	0.72	0.95	0.84
20	THieF_3wk-4root	1490.40	2400.00	0.36	0.90	0.96	0.92
21	COVIDhub-baseline	1555.90	2606.50	0.37	0.97	1.00	1.00
22	THieF_1wk-4root	1564.10	2558.40	0.29	0.87	1.00	0.98
23	THieF_4wk-4root	1595.50	2428.80	0.32	0.73	1.02	0.93
24	sarima_s1-noTransform	1763.40	2622.20	0.22	0.71	1.13	1.01
25	sarima_s1-4root	1764.90	2596.30	0.24	0.79	1.13	1.00
26	sarima_s7-noTransform	1958.00	2920.00	0.15	0.63	1.26	1.12
27	sarima_s7-4root	2198.90	3418.20	0.18	0.75	1.41	1.31

Table 2. Summary of overall model performance for the US national level, aggregated across forecast week and 1 to 4 week ahead horizons. Prediction interval (PI) coverage rates measure the fraction of times the fraction of times the 50% and 95% PIs covered the true values. A well-calibrated model should achieve close to nominal (0.5 and 0.95, respectively) coverage rates. The “rwis” and “rmae” show relative weighted interval score and relative mean absolute error, which compare each model to the baseline. The baseline is defined to have relative metrics of 1. Models that perform better than the baseline have relative WIS or MAE less than 1. The table is ordered by ascending WIS value. Thus two THieF-4wk models show the best performance out of the THieF models, meeting the baseline for both relative WIS and MAE.

	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
1	COVIDhub-4_week_ensemble	1.00	16.70	25.11	0.65	0.96	0.70	0.79
2	THieF_ensemble-train3	1.00	20.32	30.04	0.55	0.93	0.85	0.94
3	THieF_ensemble-mean	1.00	20.76	30.52	0.56	0.93	0.87	0.96
4	THieF_6wk-4root	1.00	22.50	32.97	0.49	0.89	0.94	1.03
5	sarima_s7-noTransform	1.00	23.11	31.92	0.57	0.89	0.97	1.00
6	COVIDhub-baseline	1.00	23.92	31.87	0.80	0.97	1.00	1.00
7	THieF_6wk-noTransform	1.00	25.17	36.41	0.52	0.89	1.05	1.14
8	THieF_12wk-noTransform	1.00	29.17	42.20	0.49	0.89	1.22	1.32
9	COVIDhub-4_week_ensemble	2.00	25.42	37.90	0.66	0.95	0.69	0.77
10	THieF_6wk-4root	2.00	30.10	43.94	0.52	0.91	0.81	0.89
11	THieF_ensemble-train3	2.00	30.66	44.49	0.56	0.91	0.83	0.90
12	THieF_ensemble-mean	2.00	31.04	44.80	0.58	0.91	0.84	0.90
13	THieF_6wk-noTransform	2.00	34.99	50.27	0.56	0.88	0.95	1.02
14	COVIDhub-baseline	2.00	36.95	49.48	0.80	0.96	1.00	1.00
15	sarima_s7-noTransform	2.00	37.32	50.28	0.60	0.87	1.01	1.02
16	THieF_12wk-noTransform	2.00	37.35	54.31	0.53	0.88	1.01	1.10
17	COVIDhub-4_week_ensemble	3.00	35.22	52.52	0.64	0.94	0.73	0.79
18	THieF_6wk-4root	3.00	38.98	56.36	0.52	0.90	0.80	0.85
19	THieF_ensemble-train3	3.00	41.60	59.88	0.56	0.90	0.86	0.90
20	THieF_ensemble-mean	3.00	41.85	60.07	0.58	0.90	0.86	0.91
21	THieF_6wk-noTransform	3.00	45.11	64.83	0.56	0.87	0.93	0.98
22	THieF_12wk-noTransform	3.00	46.50	68.16	0.52	0.87	0.96	1.03
23	COVIDhub-baseline	3.00	48.54	66.30	0.79	0.95	1.00	1.00
24	sarima_s7-noTransform	3.00	51.31	69.07	0.60	0.86	1.06	1.04
25	COVIDhub-4_week_ensemble	4.00	44.39	65.43	0.63	0.93	0.76	0.80
26	THieF_6wk-4root	4.00	47.53	68.48	0.52	0.88	0.81	0.84
27	THieF_ensemble-train3	4.00	51.04	73.47	0.55	0.89	0.88	0.90
28	THieF_ensemble-mean	4.00	51.11	73.39	0.56	0.88	0.88	0.90
29	THieF_6wk-noTransform	4.00	53.80	77.46	0.55	0.86	0.92	0.95
30	THieF_12wk-noTransform	4.00	54.52	80.61	0.50	0.87	0.93	0.99
31	COVIDhub-baseline	4.00	58.35	81.23	0.77	0.95	1.00	1.00
32	sarima_s7-noTransform	4.00	63.89	86.44	0.60	0.85	1.09	1.06

Table 3. Summary of model performance stratified by horizon week for the states level, aggregated across forecast week. The results from this table were used to create Figure 4 that summarized average WIS by horizon week for the US national level and the states level. The table is ordered by ascending horizon week and WIS, with THieF-4wk-4root having the lowest WIS except for the 3 week (although it is second lowest) and 4 week ahead horizon. The baseline again consistently does not have the best WIS, performing progressively worse compared to the other models for each additional week ahead horizon.

	model	horizon_wk	wis	mae	cov_50	cov_95	rwis	rmae
1	COVIDhub-4_week_ensemble	1.00	659.80	966.80	0.68	0.94	0.69	0.77
2	THieF_ensemble-train3	1.00	909.60	1348.80	0.37	0.86	0.95	1.07
3	THieF_ensemble-mean	1.00	930.90	1372.90	0.38	0.86	0.98	1.09
4	COVIDhub-baseline	1.00	952.20	1258.80	0.73	0.92	1.00	1.00
5	THieF_6wk-4root	1.00	1007.40	1419.10	0.45	0.79	1.06	1.13
6	THieF_6wk-noTransform	1.00	1028.70	1467.40	0.46	0.88	1.08	1.17
7	sarima_s7-noTransform	1.00	1143.40	1533.90	0.32	0.81	1.20	1.22
8	THieF_12wk-noTransform	1.00	1272.40	1793.60	0.40	0.88	1.34	1.42
9	COVIDhub-4_week_ensemble	2.00	1156.20	1694.70	0.66	0.92	0.72	0.80
10	THieF_6wk-4root	2.00	1413.20	2029.00	0.42	0.85	0.88	0.90
11	THieF_ensemble-train3	2.00	1417.10	2046.90	0.39	0.84	0.88	0.97
12	THieF_ensemble-mean	2.00	1436.90	2070.80	0.40	0.85	0.90	0.98
13	THieF_6wk-noTransform	2.00	1528.50	2177.00	0.47	0.86	0.95	1.03
14	COVIDhub-baseline	2.00	1604.90	2113.30	0.74	0.85	1.00	1.00
15	THieF_12wk-noTransform	2.00	1858.50	2558.30	0.38	0.85	1.16	1.23
16	sarima_s7-noTransform	2.00	1923.10	2536.30	0.41	0.82	1.20	1.20
17	COVIDhub-4_week_ensemble	3.00	1670.60	2528.80	0.60	0.90	0.75	0.84
18	THieF_6wk-4root	3.00	1867.40	2715.30	0.37	0.84	0.84	0.93
19	THieF_ensemble-train3	3.00	1979.60	2845.10	0.35	0.81	0.89	0.93
20	THieF_6wk-noTransform	3.00	1989.00	2883.60	0.46	0.85	0.90	0.90
21	THieF_ensemble-mean	3.00	1989.00	2860.10	0.39	0.83	0.90	0.93
22	COVIDhub-baseline	3.00	2217.00	2993.70	0.71	0.83	1.00	1.00
23	THieF_12wk-noTransform	3.00	2369.90	3313.80	0.37	0.86	1.07	1.13
24	sarima_s7-noTransform	3.00	2738.40	3581.40	0.46	0.80	1.24	1.20
25	COVIDhub-4_week_ensemble	4.00	2173.40	3232.70	0.56	0.89	0.79	0.83
26	THieF_6wk-4root	4.00	2330.00	3341.70	0.36	0.80	0.84	0.88
27	THieF_6wk-noTransform	4.00	2414.90	3567.40	0.43	0.85	0.88	0.94
28	THieF_ensemble-train3	4.00	2501.10	3615.30	0.32	0.80	0.91	0.90
29	THieF_ensemble-mean	4.00	2502.20	3613.60	0.38	0.80	0.91	0.93
30	COVIDhub-baseline	4.00	2761.40	3782.10	0.67	0.80	1.00	1.00
31	THieF_12wk-noTransform	4.00	2852.50	4049.10	0.31	0.86	1.03	1.07
32	sarima_s7-noTransform	4.00	3549.40	4630.40	0.45	0.76	1.28	1.22

Table 4. Summary of model performance stratified by horizon week for the US national level, aggregated across forecast week. The results from this table were used to create Figure 4 that summarized average WIS by horizon week for the US national level and the states level. The table is ordered by ascending horizon week and WIS, with THieF-4wk-4root having the lowest WIS except for the 4 week ahead horizon. The baseline consistently does not have the best WIS.

	model	horizon_wk	wave	wis	mae	cov_50	cov_95	rw
1	COVIDhub-4_week_ensemble	1.00	ba4_ba5	8.91	13.43	0.72	0.99	0.6
2	THieF_6wk-4root	1.00	ba4_ba5	9.95	14.83	0.62	0.97	0.6
3	THieF_ensemble-train3	1.00	ba4_ba5	10.48	15.62	0.67	0.99	0.7
4	THieF_ensemble-mean	1.00	ba4_ba5	11.10	16.47	0.69	0.99	0.7
5	sarima_s7-noTransform	1.00	ba4_ba5	12.81	19.00	0.68	0.98	0.8
6	THieF_6wk-noTransform	1.00	ba4_ba5	13.93	20.54	0.70	0.99	0.9
7	COVIDhub-baseline	1.00	ba4_ba5	14.49	16.59	0.92	1.00	1.0
8	THieF_12wk-noTransform	1.00	ba4_ba5	18.37	28.32	0.61	0.99	1.2
9	COVIDhub-4_week_ensemble	2.00	ba4_ba5	11.85	17.64	0.76	1.00	0.6
10	THieF_6wk-4root	2.00	ba4_ba5	13.28	19.07	0.69	0.99	0.6
11	THieF_ensemble-train3	2.00	ba4_ba5	14.15	20.34	0.74	0.99	0.7
12	THieF_ensemble-mean	2.00	ba4_ba5	14.86	21.20	0.75	0.99	0.7
13	sarima_s7-noTransform	2.00	ba4_ba5	17.46	25.07	0.75	0.99	0.8
14	THieF_6wk-noTransform	2.00	ba4_ba5	17.92	25.69	0.77	1.00	0.9
15	COVIDhub-baseline	2.00	ba4_ba5	19.69	21.67	0.97	1.00	1.0
16	THieF_12wk-noTransform	2.00	ba4_ba5	21.52	32.82	0.68	1.00	1.0
17	COVIDhub-4_week_ensemble	3.00	ba4_ba5	15.22	22.86	0.76	0.99	0.6
18	THieF_6wk-4root	3.00	ba4_ba5	16.82	23.30	0.72	0.99	0.7
19	THieF_ensemble-train3	3.00	ba4_ba5	17.95	25.29	0.76	1.00	0.7
20	THieF_ensemble-mean	3.00	ba4_ba5	18.68	26.22	0.78	1.00	0.7
21	THieF_6wk-noTransform	3.00	ba4_ba5	21.59	31.20	0.78	1.00	0.9
22	sarima_s7-noTransform	3.00	ba4_ba5	22.06	31.34	0.78	0.99	0.9
23	COVIDhub-baseline	3.00	ba4_ba5	23.93	27.49	0.97	1.00	1.0
24	THieF_12wk-noTransform	3.00	ba4_ba5	25.38	39.28	0.70	1.00	1.0
25	COVIDhub-4_week_ensemble	4.00	ba4_ba5	18.26	27.51	0.76	0.99	0.6
26	THieF_6wk-4root	4.00	ba4_ba5	19.92	26.57	0.75	1.00	0.7
27	THieF_ensemble-train3	4.00	ba4_ba5	21.50	29.63	0.78	1.00	0.7
28	THieF_ensemble-mean	4.00	ba4_ba5	22.13	30.41	0.79	1.00	0.8
29	THieF_6wk-noTransform	4.00	ba4_ba5	24.17	34.33	0.80	1.00	0.8
30	sarima_s7-noTransform	4.00	ba4_ba5	25.95	36.72	0.80	0.99	0.9
31	COVIDhub-baseline	4.00	ba4_ba5	27.46	32.80	0.97	1.00	1.0
32	THieF_12wk-noTransform	4.00	ba4_ba5	29.06	45.41	0.69	1.00	1.0
33	sarima_s7-noTransform	1.00	delta	10.38	14.69	0.70	0.97	0.3
34	THieF_ensemble-mean	1.00	delta	11.60	18.76	0.64	0.99	0.3
35	THieF_ensemble-train3	1.00	delta	11.87	19.30	0.60	0.99	0.3
36	THieF_6wk-4root	1.00	delta	14.16	22.68	0.51	0.95	0.4
37	THieF_12wk-noTransform	1.00	delta	14.89	23.52	0.52	0.95	0.4
38	THieF_6wk-noTransform	1.00	delta	15.57	25.83	0.51	0.97	0.4
39	sarima_s7-noTransform	2.00	delta	14.64	20.80	0.69	0.98	0.2
40	THieF_ensemble-mean	2.00	delta	16.33	26.27	0.65	0.99	0.3
41	THieF_ensemble-train3	2.00	delta	16.52	26.76	0.64	0.98	0.3
42	THieF_6wk-4root	2.00	delta	17.99	29.88	0.54	0.98	0.3
43	THieF_12wk-noTransform	2.00	delta	19.35	30.10	0.60	0.95	0.3
44	THieF_6wk-noTransform	2.00	delta	19.58	32.20	0.62	0.97	0.3
45	sarima_s7-noTransform	3.00	delta	23.20	35.71	0.60	0.97	0.3
46	THieF_ensemble-mean	3.00	delta	25.61	41.49	0.58	0.97	0.3
47	THieF_ensemble-train3	3.00	delta	25.77	41.94	0.57	0.98	0.3
48	THieF_6wk-4root	3.00	delta	27.03	44.49	0.52	0.99	0.3

	model	horizon_wk	wave	wis	mae	cov_50	cov_95	
1	COVIDhub-4_week_ensemble	1.00	ba4_ba5	277.80	378.70	0.89	1.00	
2	THieF_6wk-4root	1.00	ba4_ba5	350.40	542.90	0.60	0.99	
3	THieF_ensemble-train3	1.00	ba4_ba5	399.10	685.30	0.46	1.00	
4	THieF_ensemble-mean	1.00	ba4_ba5	433.60	740.30	0.49	1.00	
5	COVIDhub-baseline	1.00	ba4_ba5	450.80	486.90	0.92	1.00	
6	THieF_6wk-noTransform	1.00	ba4_ba5	516.50	822.50	0.60	1.00	
7	sarima_s7-noTransform	1.00	ba4_ba5	517.80	828.60	0.41	0.99	
8	THieF_12wk-noTransform	1.00	ba4_ba5	790.70	1319.20	0.35	1.00	
9	COVIDhub-4_week_ensemble	2.00	ba4_ba5	411.40	572.50	0.85	1.00	
10	THieF_6wk-4root	2.00	ba4_ba5	518.80	813.00	0.61	1.00	
11	THieF_ensemble-train3	2.00	ba4_ba5	578.60	966.60	0.49	1.00	
12	THieF_ensemble-mean	2.00	ba4_ba5	621.80	1030.30	0.50	1.00	
13	COVIDhub-baseline	2.00	ba4_ba5	654.20	741.00	0.98	1.00	
14	THieF_6wk-noTransform	2.00	ba4_ba5	727.80	1145.90	0.59	1.00	
15	sarima_s7-noTransform	2.00	ba4_ba5	729.70	1169.60	0.53	1.00	
16	THieF_12wk-noTransform	2.00	ba4_ba5	1012.60	1677.50	0.38	1.00	
17	COVIDhub-4_week_ensemble	3.00	ba4_ba5	563.30	824.70	0.77	1.00	
18	THieF_6wk-4root	3.00	ba4_ba5	683.90	1069.10	0.59	1.00	
19	THieF_ensemble-train3	3.00	ba4_ba5	784.00	1311.50	0.45	1.00	
20	COVIDhub-baseline	3.00	ba4_ba5	825.00	1038.90	1.00	1.00	
21	THieF_ensemble-mean	3.00	ba4_ba5	825.00	1371.30	0.50	1.00	
22	THieF_6wk-noTransform	3.00	ba4_ba5	911.00	1464.40	0.59	1.00	
23	sarima_s7-noTransform	3.00	ba4_ba5	953.50	1514.20	0.57	1.00	
24	THieF_12wk-noTransform	3.00	ba4_ba5	1248.70	2097.10	0.38	1.00	
25	COVIDhub-4_week_ensemble	4.00	ba4_ba5	698.90	1034.30	0.77	1.00	
26	THieF_6wk-4root	4.00	ba4_ba5	801.90	1210.30	0.64	1.00	
27	THieF_ensemble-train3	4.00	ba4_ba5	969.80	1585.10	0.48	1.00	
28	COVIDhub-baseline	4.00	ba4_ba5	972.00	1320.50	1.00	1.00	
29	THieF_ensemble-mean	4.00	ba4_ba5	1010.80	1651.70	0.53	1.00	
30	THieF_6wk-noTransform	4.00	ba4_ba5	1034.00	1631.70	0.61	1.00	
31	sarima_s7-noTransform	4.00	ba4_ba5	1169.60	1821.80	0.60	1.00	
32	THieF_12wk-noTransform	4.00	ba4_ba5	1501.80	2581.80	0.34	1.00	
33	THieF_ensemble-mean	1.00	delta	239.10	305.40	0.86	1.00	
34	THieF_ensemble-train3	1.00	delta	247.90	326.80	0.86	1.00	
35	sarima_s7-noTransform	1.00	delta	270.90	454.30	0.29	1.00	
36	THieF_6wk-4root	1.00	delta	327.60	536.40	0.71	1.00	
37	THieF_6wk-noTransform	1.00	delta	332.50	563.70	0.43	1.00	
38	THieF_12wk-noTransform	1.00	delta	381.80	523.00	1.00	1.00	
39	THieF_6wk-noTransform	2.00	delta	286.50	260.10	1.00	1.00	
40	sarima_s7-noTransform	2.00	delta	290.70	240.00	1.00	1.00	
41	THieF_ensemble-mean	2.00	delta	313.60	215.30	1.00	1.00	
42	THieF_ensemble-train3	2.00	delta	316.20	207.60	1.00	1.00	
43	THieF_6wk-4root	2.00	delta	324.40	280.40	1.00	1.00	
44	THieF_12wk-noTransform	2.00	delta	487.90	404.70	1.00	1.00	
45	THieF_6wk-noTransform	3.00	delta	396.00	511.10	0.86	1.00	
46	sarima_s7-noTransform	44 3.00	delta	486.30	701.80	1.00	1.00	
47	THieF_ensemble-mean	3.00	delta	509.60	800.20	1.00	1.00	
48	THieF_ensemble-train3	3.00	delta	509.60	797.20	1.00	1.00	

BIBLIOGRAPHY

- [1] Athanasopoulos, G., Hyndman, R.J., Kourentzes, N., and Petropoulos, F. Forecasting with temporal hierarchies. *European Journal of Operational Research* 262, 1 (October 2017), 60–74.
- [2] COVID-19 Forecast Hub. Ensemble model. <https://covid19forecasthub.org/doc/ensemble/>. Accessed: 2023-03-15.
- [3] COVID-19 Forecast Hub. Forecast evaluations. <https://covid19forecasthub.org/eval-reports/>. Accessed: 2023-03-18.
- [4] Cramer, Estee Y, Huang, Yuxin, Wang, Yijin, Ray, Evan L, Cornell, Matthew, Bracher, Johannes, Brennen, Andrea, Castro Rivadeneira, Alvaro J, Gerding, Aaron, House, Katie, Jayawardena, Dasuni, Kanji, Abdul H, Khandelwal, Ayush, Le, Khoa, Niemi, Jarad, Stark, Ariane, Shah, Apurv, Wattanachit, Nutch, Zorn, Martha W, Reich, Nicholas G, and Consortium, US COVID-19 Forecast Hub. The united states covid-19 forecast hub dataset. *medRxiv* (2021).
- [5] Cramer, Estee Y., Ray, Evan L., Lopez, Velma K., Bracher, Johannes, Brennen, Andrea, Rivadeneira, Alvaro J. Castro, Gerding, Aaron, Gneiting, Tilmann, Katie H. House, Yuxin Huang, Dasuni Jayawardena, Kanji, Abdul H., Khandelwal, Ayush, Le, Khoa, Mühlemann, Anja, Niemi, Jarad, Shah, Apurv, Stark, Ariane, Wang, Yijin, Wattanachit, Nutch, ..., and Reich, Nicholas G. Evaluation of individual and ensemble probabilistic forecasts of covid-19 mortality in the us. *Proceedings of the National Academy of Sciences* 119, 15 (April 2022), e2113561119.
- [6] Donovan, Doug. Johns hopkins covid-19 data hub ends after three years. *The Hub* (March 2023).
- [7] Gibson, Graham C., Reich, Nicholas G., and Sheldon, Daniel. Real-time mechanistic bayesian forecasts of covid-19 mortality. *medRxiv : the preprint server for health sciences* (December 2020).
- [8] Kourentzes, Nikos. Improving organizational forecasts and decisions with hierarchical forecasting. <https://youtu.be/dpASMg2jzQY>, 2021. Accessed: 2022-04-06.
- [9] Lauer, Stephen A., Brown, Alexandria C., and Reich, Nicholas G. Infectious disease forecasting for public health, 2020.

- [10] Ray, Evan L., Brooks, Logan C., Bien, Jacob, Biggerstaff, Matthew, Bosse, Nikos I., Bracher, Johannes, Cramer, Estee Y., Funk, Sebastian, Gerding, Aaron, Johansson, Michael A., Rumack, Aaron, Wang, Yijin, Zorn, Martha, Tibshirani, Ryan J., and Reich, Nicholas G. Comparing trained and untrained probabilistic ensemble forecasts of covid-19 cases and deaths in the united states. *International Journal of Forecasting* (July 2022).
- [11] Reich, Nicholas G., McGowan, Craig J., Yamana, Teresa K., Tushar, Abhinav, Ray, Evan L., Osthus, Dave, Kandula, Sasikiran, Brooks, Logan C., Crawford-Crudell, Willow, Gibson, Graham Casey, Moore, Evan, Silva, Rebecca, Biggerstaff, Matthew, Johansson, Michael A., Rosenfeld, Roni, and Shaman, Jeffrey. Accuracy of real-time multi-model ensemble forecasts for seasonal influenza in the u.s. *PLoS Computational Biology* 15, 11 (November 2019), e1007486.
- [12] Reich, Nicholas G., Tibshirani, Ryan J., Ray, Evan L., and Rosenfeld, Roni. On the predictability of covid-19. <https://forecasters.org/blog/2021/09/28/on-the-predictability-of-covid-19/>, September 2021. Accessed: 2022-04-06.
- [13] Reich, Nicholas G., Wang, Yijin, Burns, Meagan, Ergas, Rosa, Cramer, Estee Y., and Ray, Evan L. Assessing the utility of covid-19 case reports as a leading indicator for hospitalization forecasting in the united states. *medRxiv* (March 2023).
- [14] U.S. Department of Health and Human Services. Covid-19 reported patient impact and hospital capacity by state timeseries. <https://healthdata.gov/Hospital/COVID-19-Reported-Patient-Impact-and-Hospital-Capa/g62h-syeh>. Accessed: 2023-03-15.
- [15] Zoltar. About. <https://zoltardata.com/about>. Accessed: 2023-03-18.